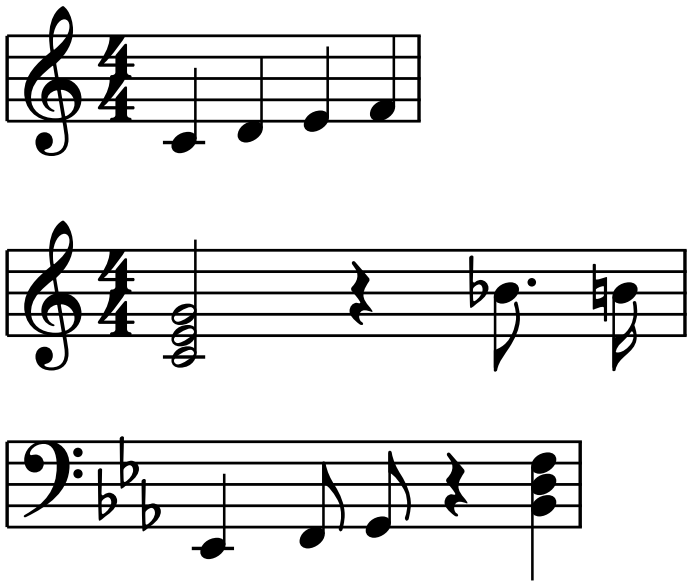typed-scores Visual Regression Suite

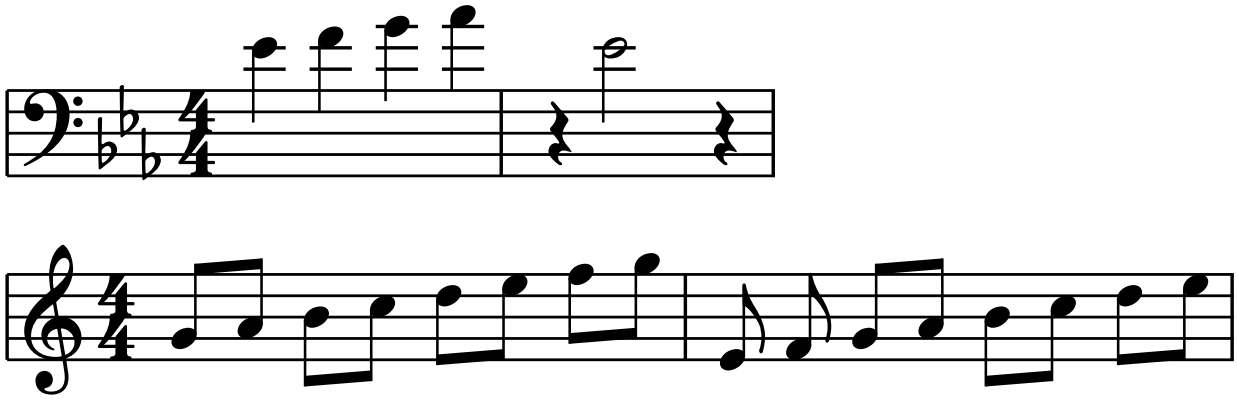
`bar()` convenience wrapper

The public quick-bar helper uses the same layout and validation pipeline as a one-bar single-staff score.
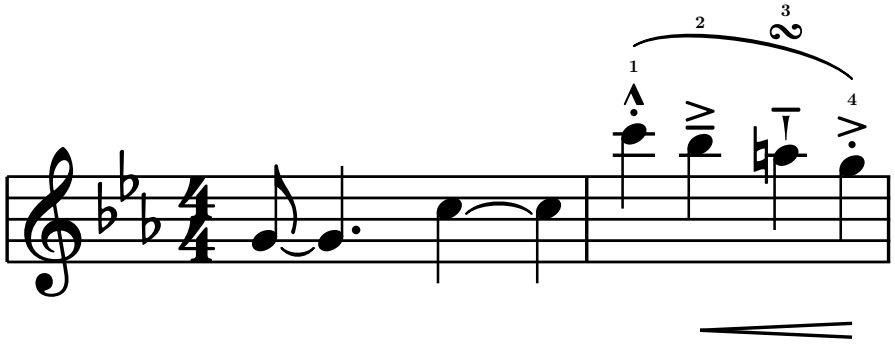


Single-staff score bars

The implicit single staff is expressed only with `notes:`.

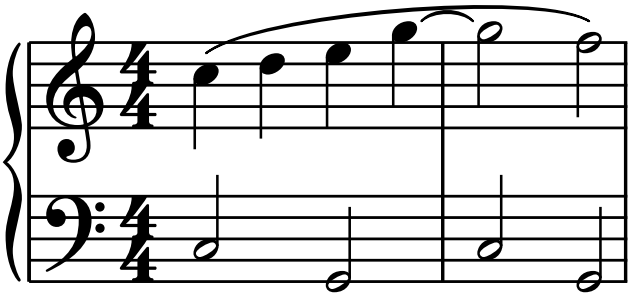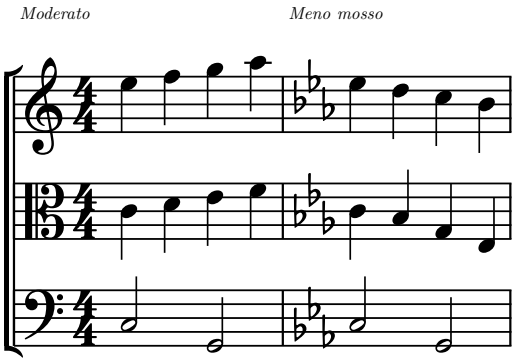


Ties, slurs, and expressive marks
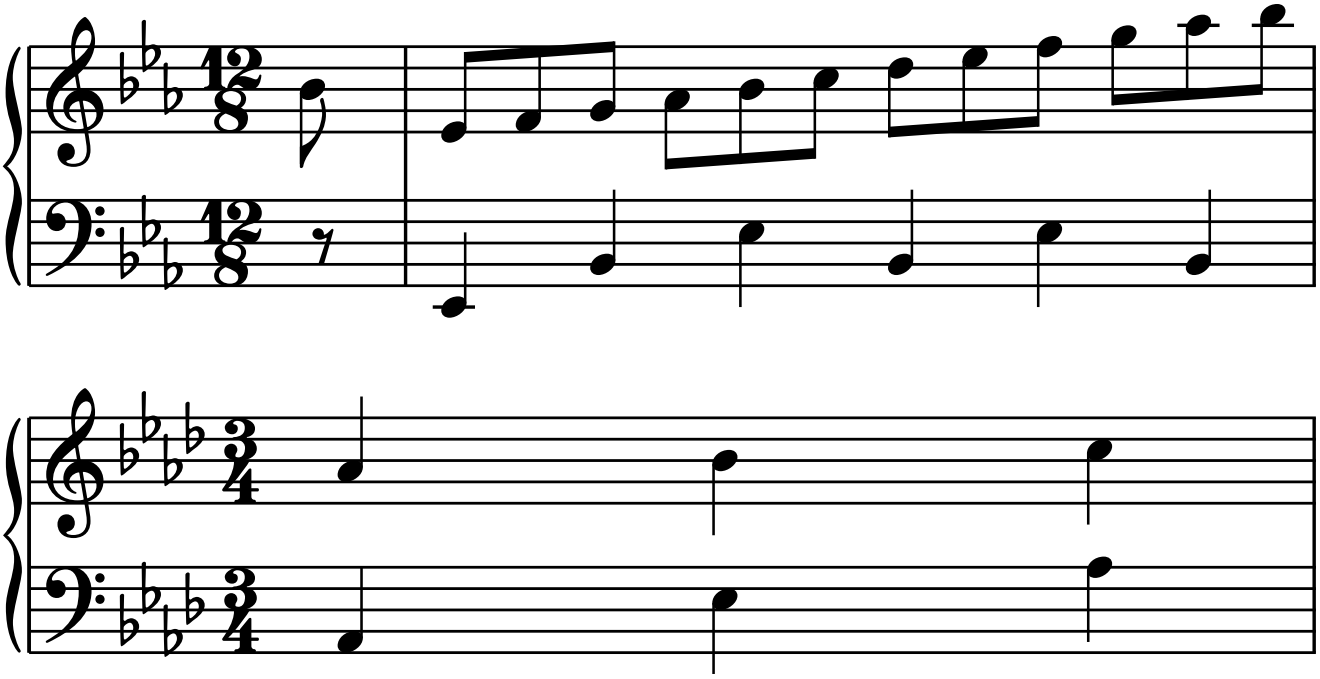


Canonical multi-staff bars

Staves are declared once; every bar supplies each staff ID.

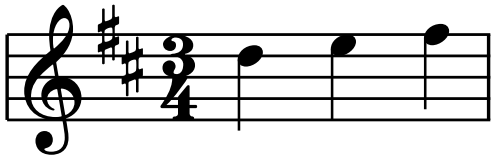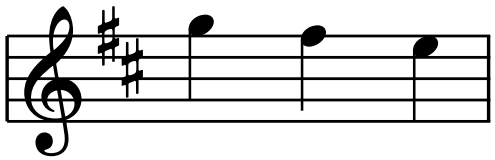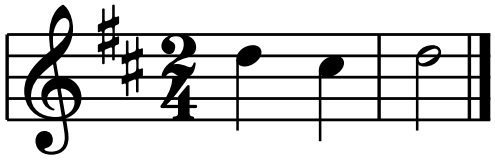




Pickups and signature changes



Each system restates its clef and key, but a time signature appears only where the meter is set or changes. The second system continues in 3/4 without restating it, and the change to 2/4 opens the third.

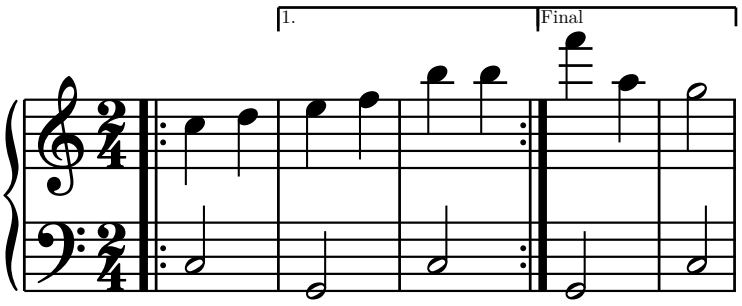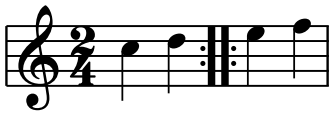






Repeat barlines and volta endings
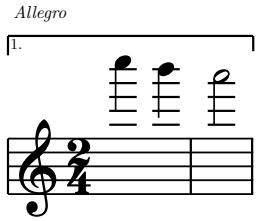
Repeat signs live on the left or right edge of a bar. Labeled endings span bars and pair their `start` and `stop` markers.



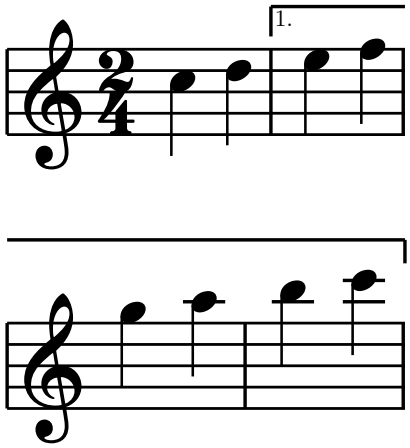
Combined end/start repeats keep their two heavy strokes distinct:



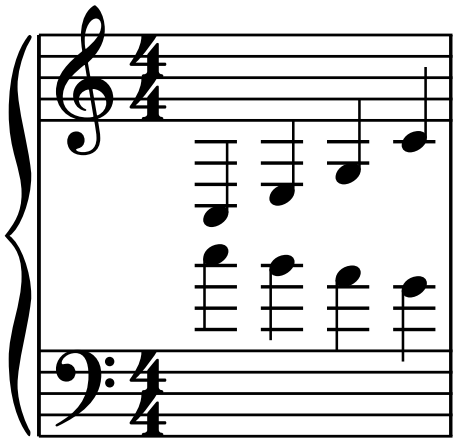
The bracket clears extreme notes and remains proportionate at small scales:



Volta brackets continue when a system wraps:



Extreme-register and spacing stress
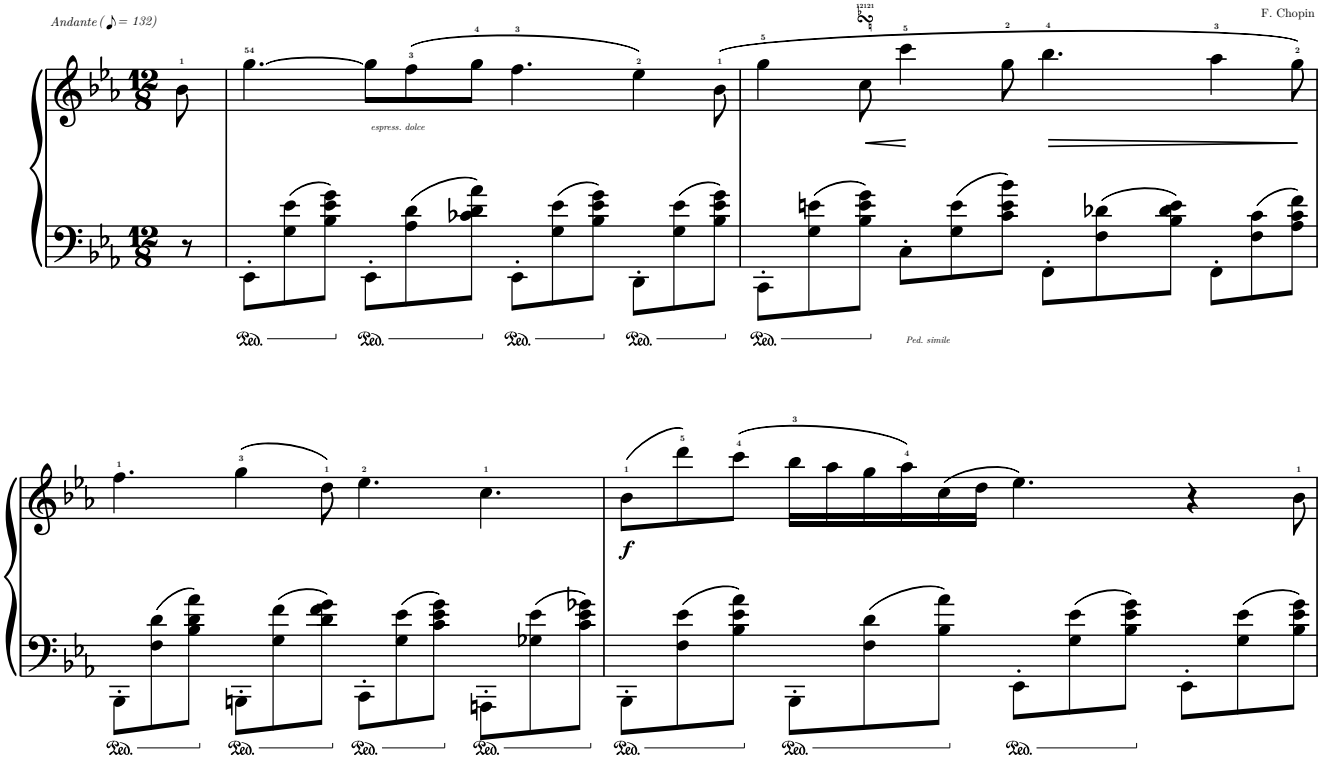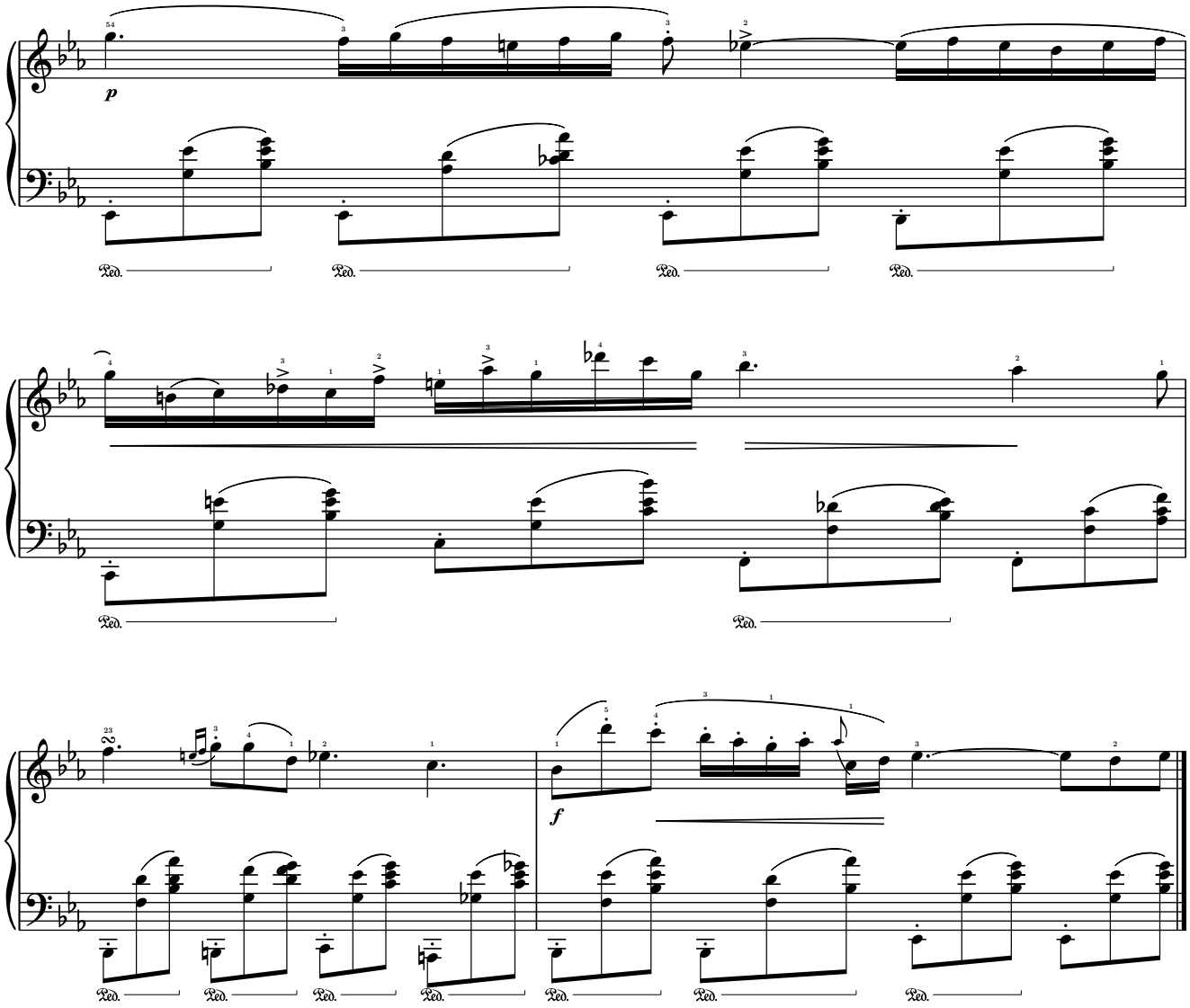


Expected compile errors

These source snippets are intentionally not compiled:

```
#score(bars: ((notes: "C5:q"),), time: "4/4")
#score(
  staves: (upper: (clef: "treble")),
  bars: ((lower: "C5:w"),),
)
#score(
  staves: (upper: (clef: "treble"), lower: (clef: "bass")),
  bars: ((upper: "C5:w"),),
)
#score(
  clef: "treble",
  bars: ((notes: "C5:q[s1(] D5:q E5:q F5:q"),),
)
```

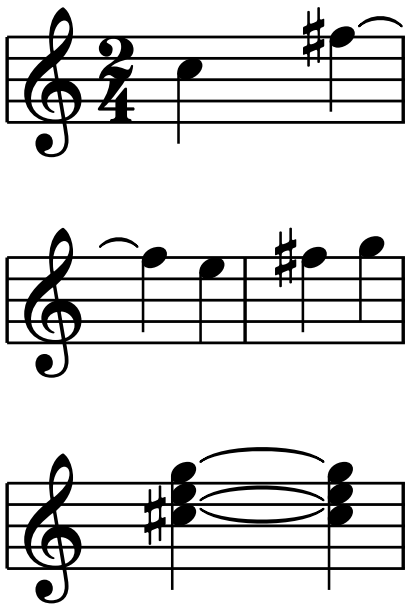
Release gate: Chopin Op. 9 No. 2
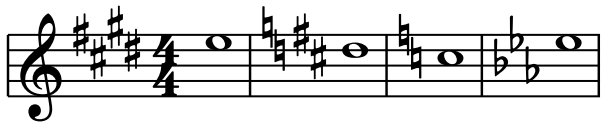




Tie semantics and wrapped continuations
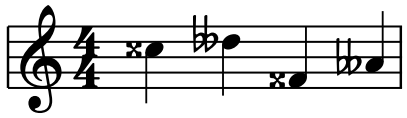
The first tie crosses a system boundary. Tied continuation accidentals are not repeated, while a later re-articulation shows the accidental normally.



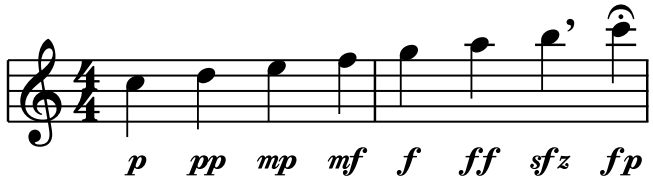
Key cancellation and double accidentals

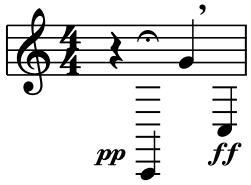
Naturals cancel only the alterations removed from the old key, and appear before the replacement signature.
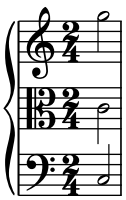




Dynamics, holds, and pauses
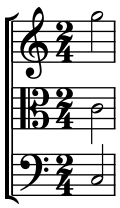




Staff grouping control
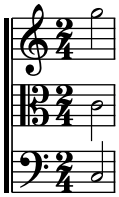
Brace
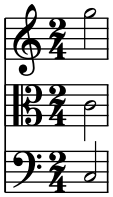


Bracket



Line
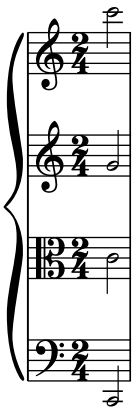


None
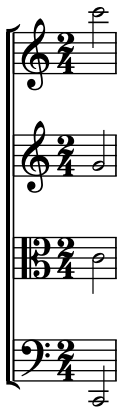


Group symbol geometry stress

The Bravura terminals and proportional brace outlines remain complete across a tall four-staff group. Every vertical stem starts at the lowest staff line and stops at the highest staff line.
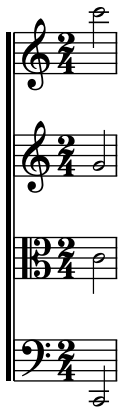
Tall brace



Bravura bracket
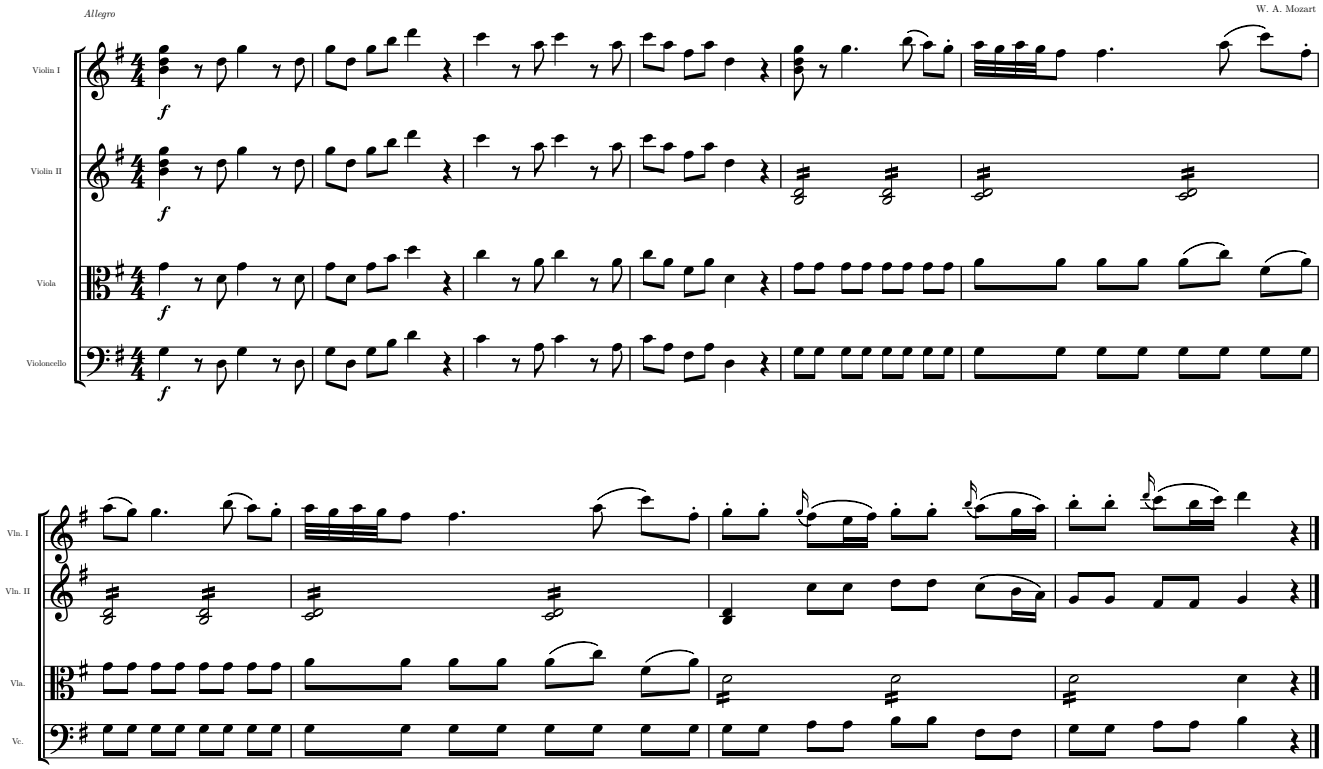


Heavy line



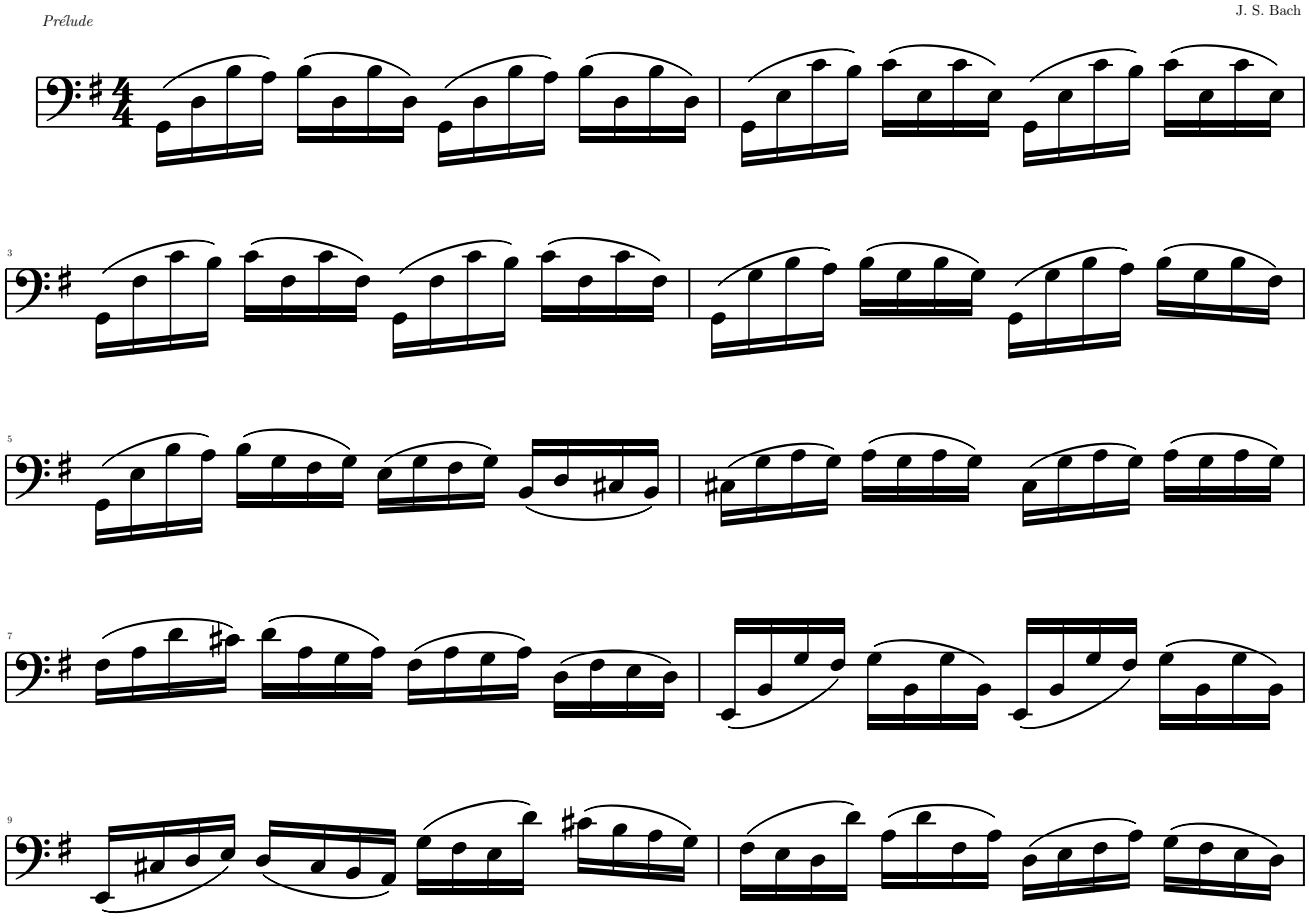
Famous-score fixture gallery

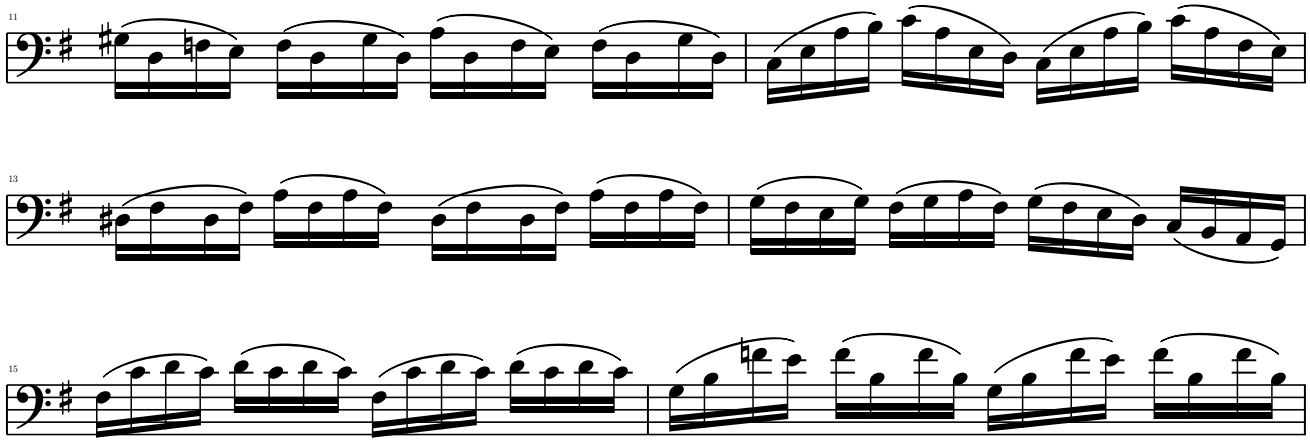
These public-domain excerpts exercise the release examples at the same time as the notation engine: quartet grouping, dense solo beaming, a transposing single staff, and a chromatic piano pickup.

Mozart · Eine kleine Nachtmusik, K. 525
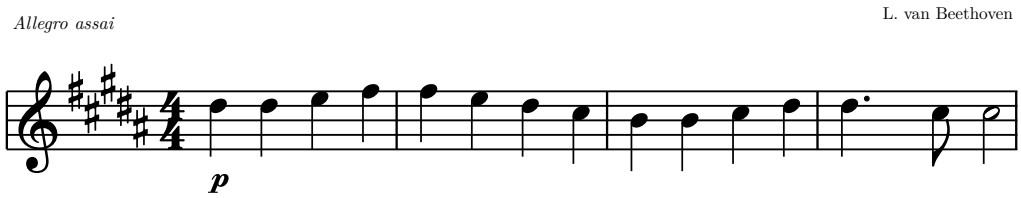


Bach · Cello Suite No. 1, BWV 1007

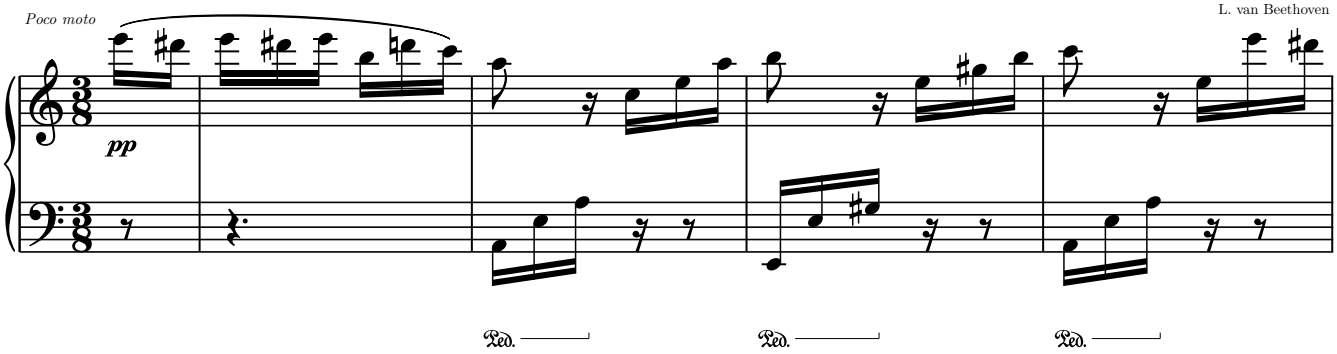




Beethoven · Ode to Joy for E-flat alto saxophone
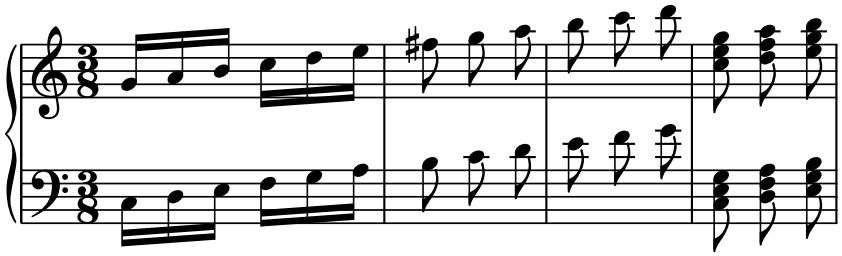


Beethoven · Für Elise, WoO 59
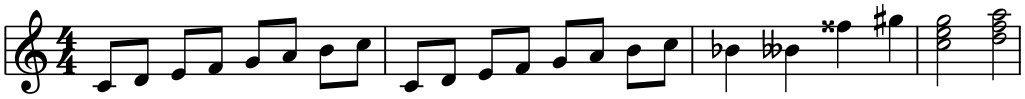


Relative notes and chords, inherited durations, and local 3+3 beams

The key is intentionally omitted, so this uses the C-major default. Each staff begins in its clef's default register, keeps its own pitch and duration anchors across barlines, and groups six sixteenths as 3+3 with local - joins and / breaks. The last bar resolves every chord from its first written pitch and then resolves the remaining chord pitches in written order.
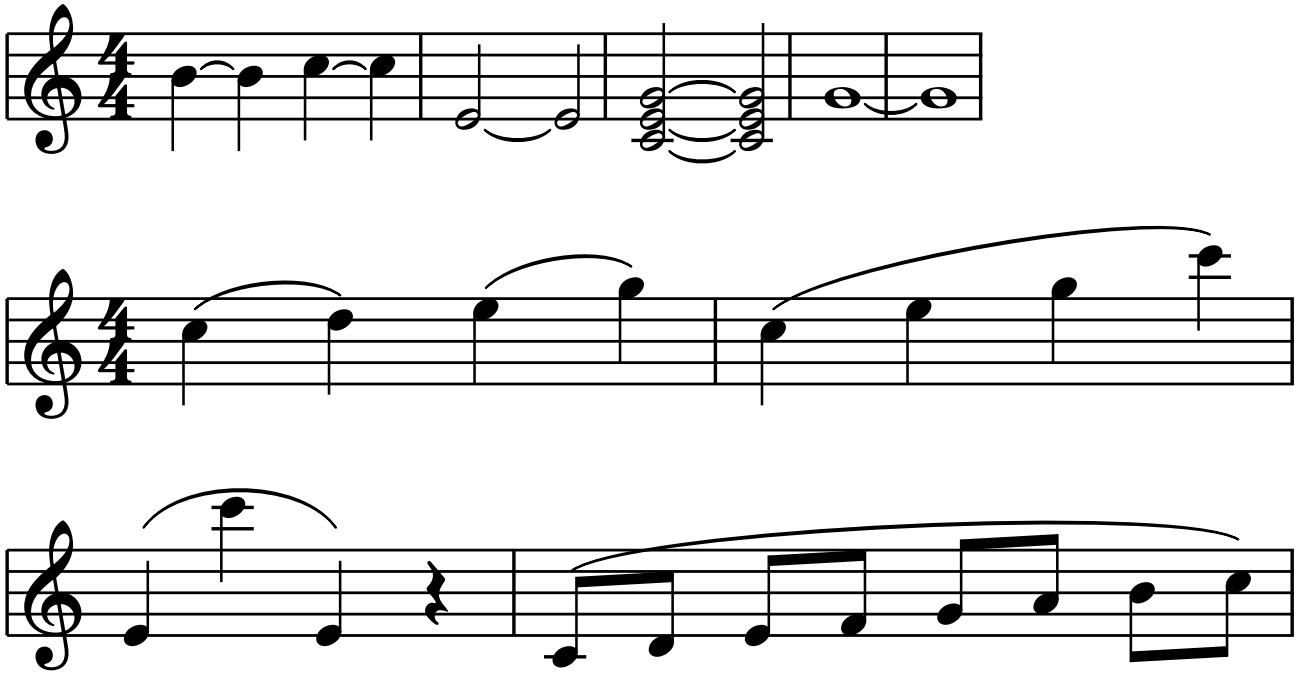


Lowercase, uppercase, and compact duration equivalence

Pitch-letter capitalization has no musical meaning. The first two bars render the same scale using compact `ce` and explicit `c:e`; mixed case is accepted. The last bars exercise lowercase flats and double flats, a compact double sharp, and mixed-case relative chord entry.



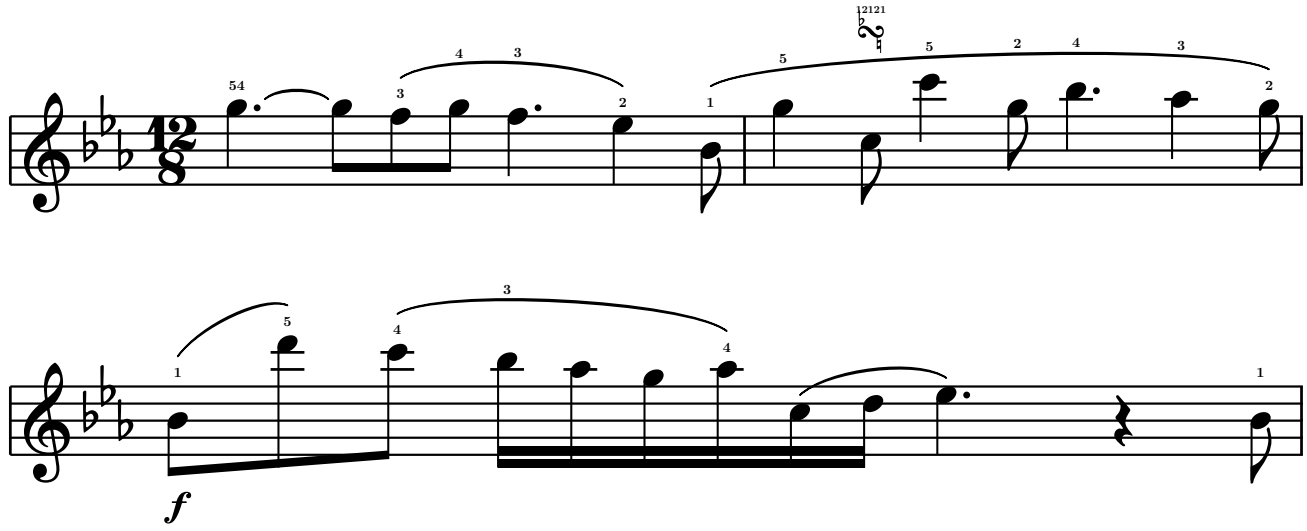
Engraved tie and slur bows

Bows are cubic Beziers whose height saturates with the span and whose shoulders flatten long arcs. Ties center inside a staff space and keep their tips and apex clear of staff lines; ties on a line borrow the adjacent space. Slurs pick endpoint raises by scored candidates: the high middle note lifts both endpoints instead of ballooning the arch, the rising line tilts the bow with the melody, and the long phrase goes flat on top with rounded shoulders.



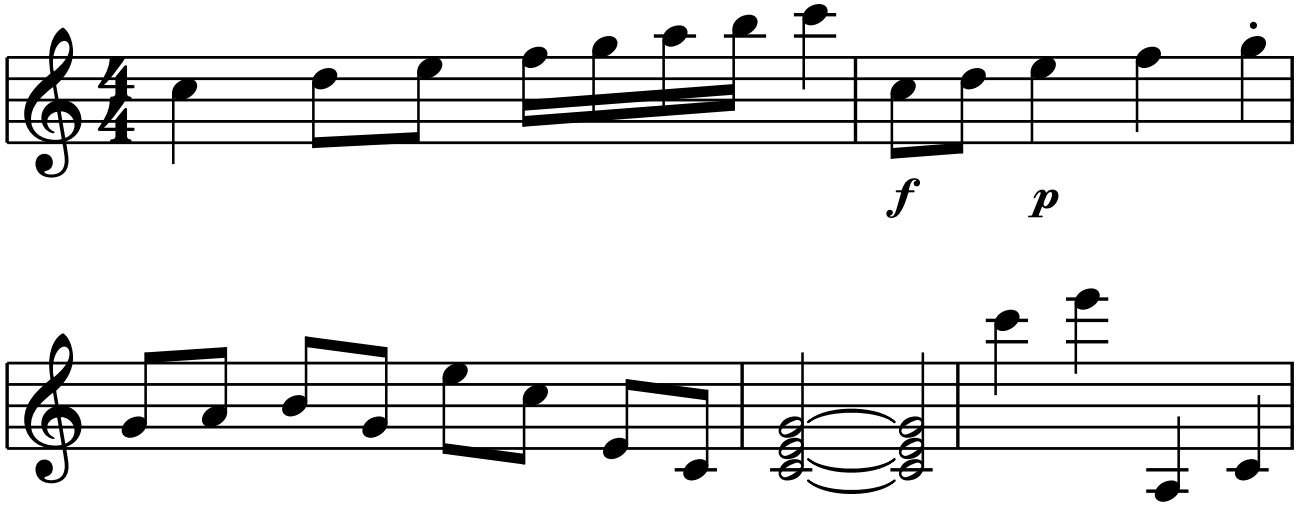
Slur, fingering, and dynamic interaction

Ties leaving dotted notes clear the augmentation dots, and ties outside the staff float past the notehead instead of centering on it. Slur tips sit above their own note's fingering, while fingerings and turn ornaments inside a slur move above the finished bow. A slur with nothing inside it keeps its tips at the notes no matter how steep the interval, and dynamics drop below stems and beams that reach under the staff.
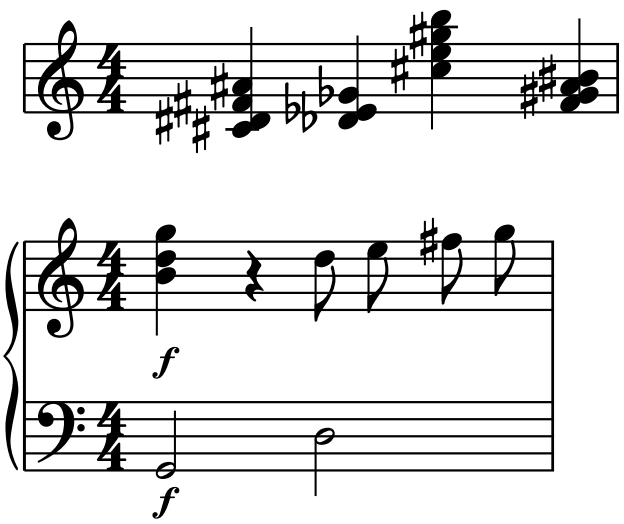


Classical spacing, beam quanting, and aligned dynamics

Horizontal space grows with the logarithm of duration from the bar's shortest note, so mixed rhythms breathe like engraved plate work. Beam slants follow the outer interval in quarter-space steps, go flat over concave contours, and their ends sit on, straddle, or hang from staff lines. Dynamics in one voice share a system baseline, staccato dots keep to staff spaces, ledger-line stems reach the middle line, and chord ties curve outward from the chord.
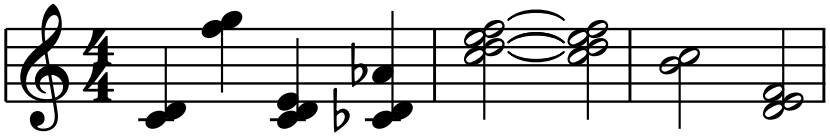


Accidental columns, systemic barline, and computed staff gaps

Chord accidentals pack into columns from the outside in - topmost nearest the heads, then bottommost - so clusters never overlap. Multi-staff systems open with a barline joining the staves, and with no explicit `staff-gap` the gaps come from the actual ink: the forte under the upper staff's down-stem chord pushes the staves apart just enough.



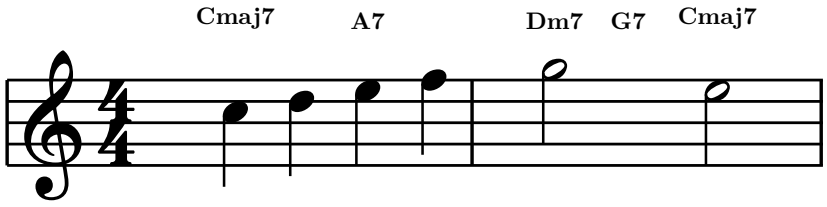
Seconds in chords

Notes a second (or unison) apart cannot share a notehead column: the upper note of each clashing pair moves to the right of the stem while the lower keeps the left, alternating through longer clusters, with accidentals, dots, ties, and spacing making room for the widened chord.
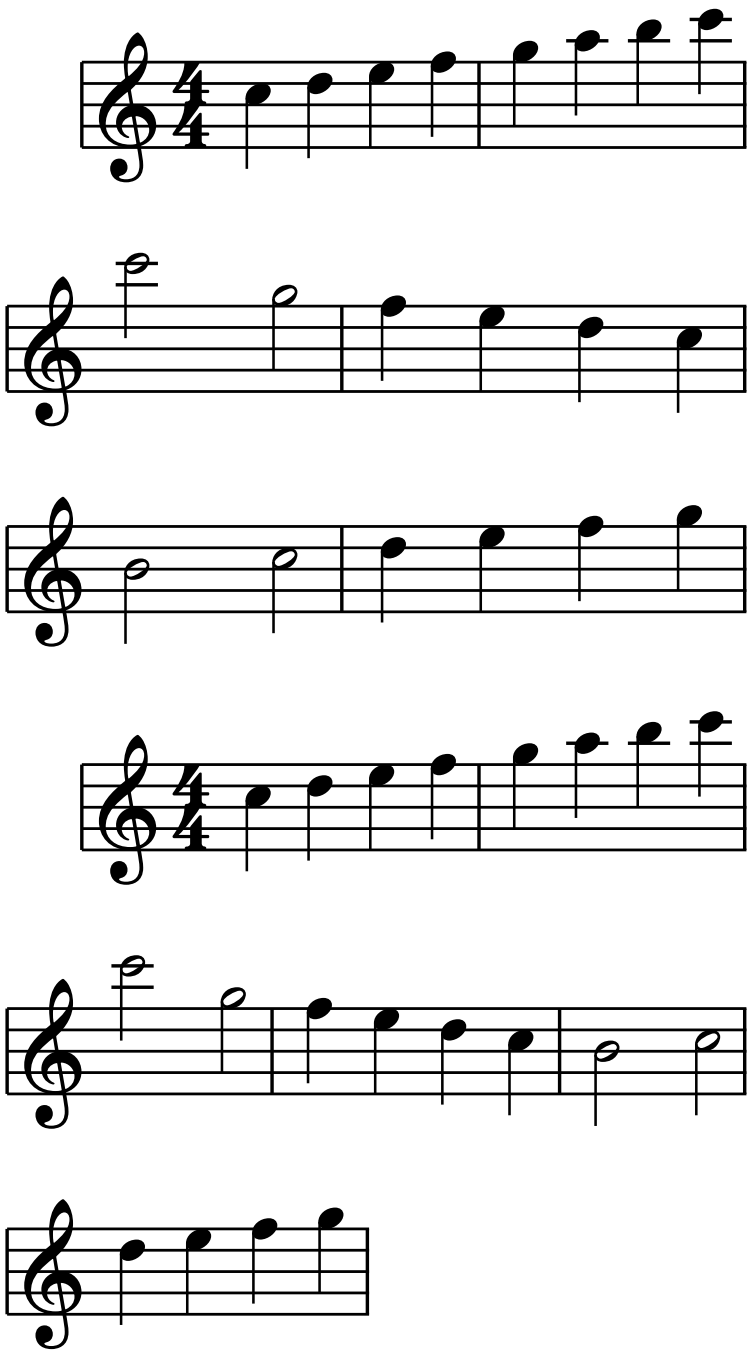


Harmony symbols

Harmony is an independent duration-bearing timeline in each bar. Symbols remain above the top staff and are centered on the onset where they become active.
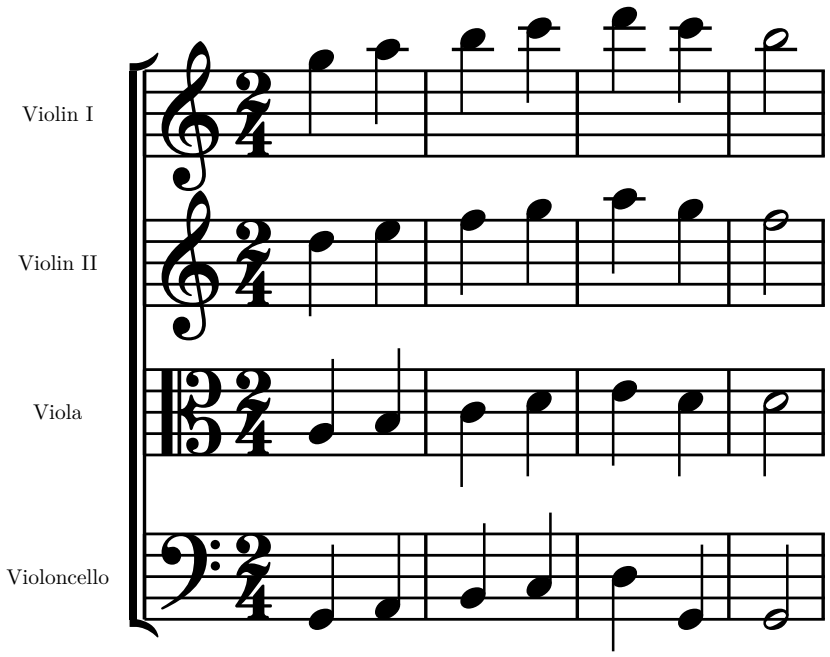


Justified systems and indentation

Multi-system scores use LilyPond-style justification by default: timed gaps stretch to the requested width, while the opening system may reserve a distinct indent. The final example opts into a natural-width final system.



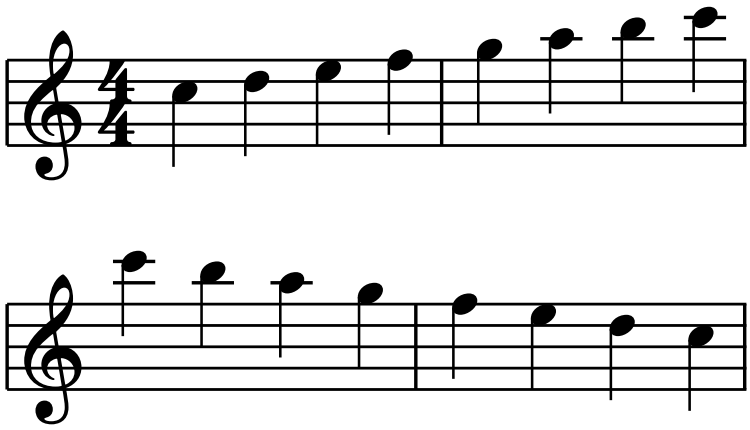
Staff labels

Full staff labels occupy the opening-system name column; optional short labels occupy the corresponding column in later systems. Each label centers on its own staff, independently of the group symbol.



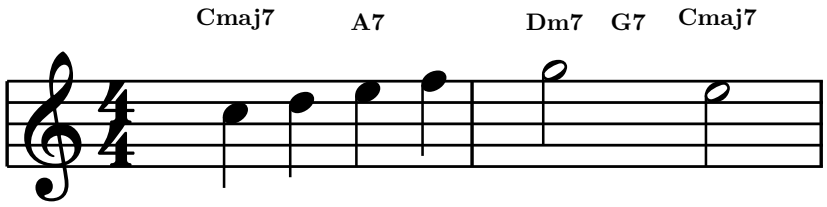
Global system balancing

Line breaking evaluates all complete-bar partitions instead of taking the first overflow. This evenly spaced passage chooses two similarly dense systems rather than a compressed three-bar system followed by one over-stretched bar.



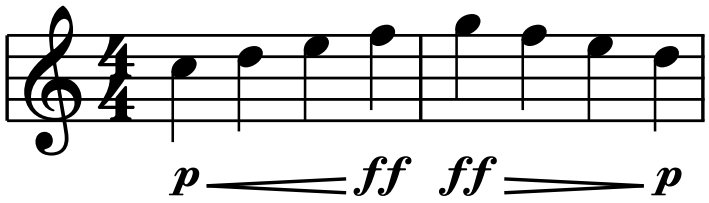
Harmony onset anchoring

Chord-symbol duration determines the next change, not the horizontal anchor. Every symbol centers on its own onset: a half-note Cmaj7 centers on the first quarter-note onset, while the following A7 centers on the third.



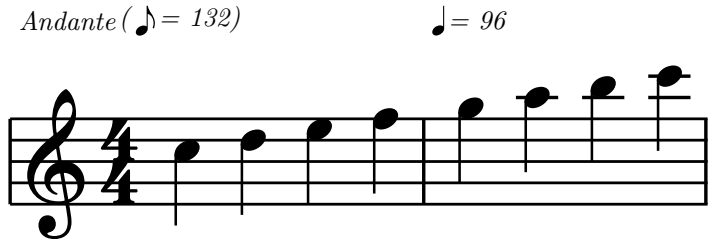
Dynamics and hairpin clearance

Dynamics and hairpins share one baseline. A wedge shortens before its ending dynamic and begins after a dynamic at its opening event, keeping both labels clear at normal and compact spacing.



Semantic metronome marks
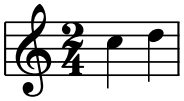
Tempo dictionaries use named beat values, so users never need to paste an eighth-note character. The renderer supplies the appropriate Bravura glyph.
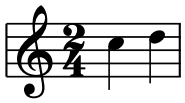


Metronome glyph set

Every supported tempo beat is tested beside tempo text and parentheses. Each Bravura glyph maintains clearance from the surrounding mark.
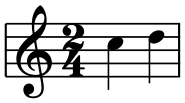
Whole (𝅝 = 60)
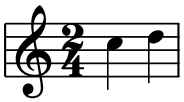


Half (𝅗𝅥 = 72)
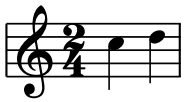


Quarter (♩ = 96)
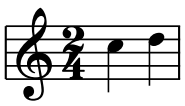


Eighth (♪ = 132)



Sixteenth (𝅘𝅥𝅯 = 168)

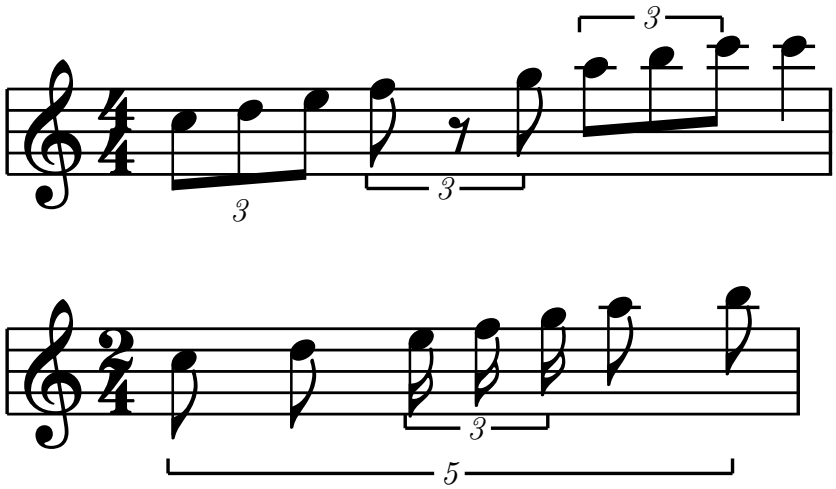


Thirty-second (𝅘𝅥𝅰 = 216)

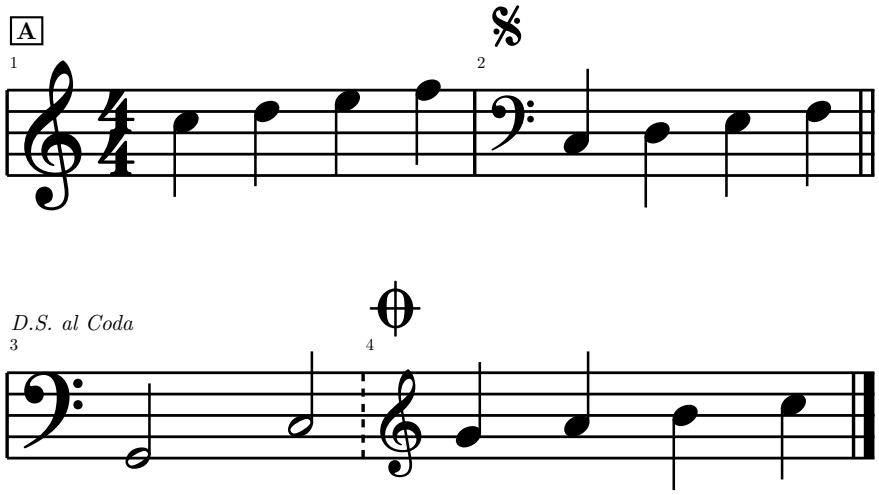


Tuplets

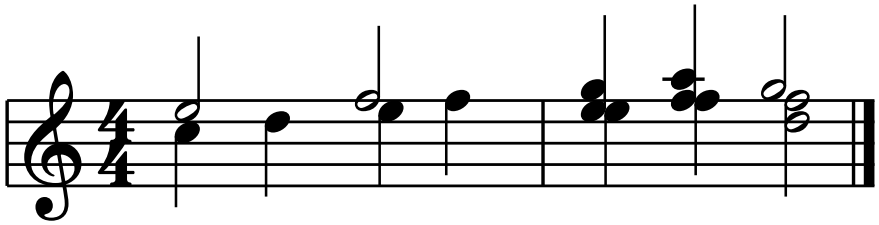
Tuplets remain inline in the note string. A complete beam suppresses the bracket, while rests or an explicit option retain it.
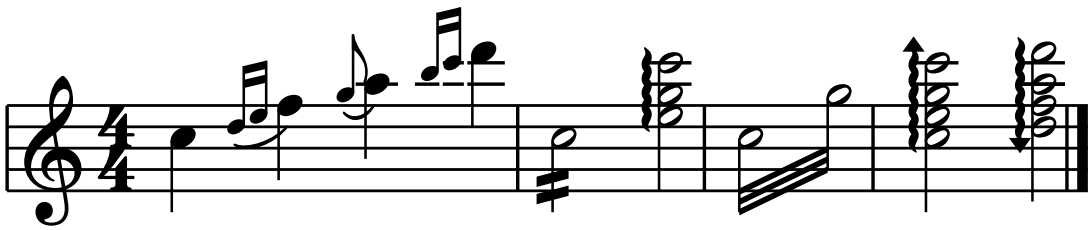


Clef changes, barlines, and navigation



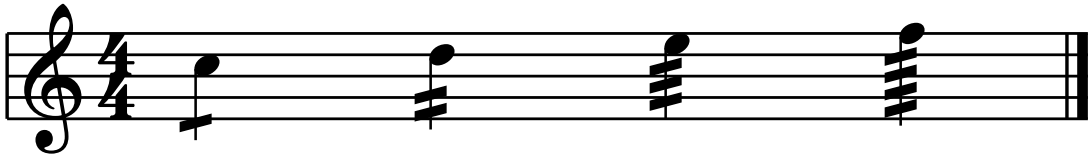
Polyphonic voices



Grace notes, tremolos, and arpeggios



Single-event tremolo subdivisions 8 through 64 select one through four strokes independently of the written note value. Quarter notes below exercise both stem directions and verify that every cluster remains clear of its notehead.
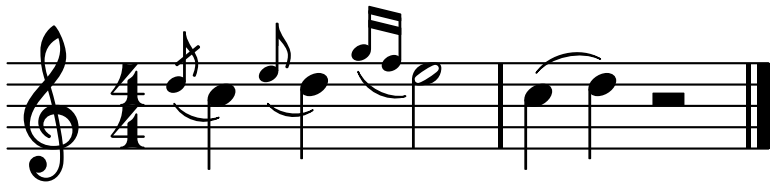


LilyPond A/B: grace notation

Each committed LilyPond 2.26.0 reference is followed by live `typed-scores` output using only the public API. The reference SVG is sized so one LilyPond staff space equals the renderer's 8pt staff space. This compares geometry at a common engraving scale even though LilyPond uses Emmentaler and `typed-scores` uses Bravura.

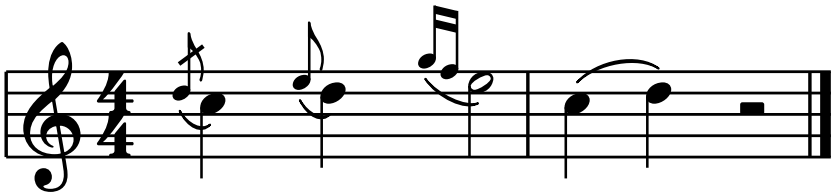
Documented grace forms

The tapered grace-slur tips retain LilyPond's visible clearance from both the small grace head and the full-size principal head. The final ordinary slur confirms that grace slurs retain the normal, unscaled bow weight.

LilyPond 2.26.0

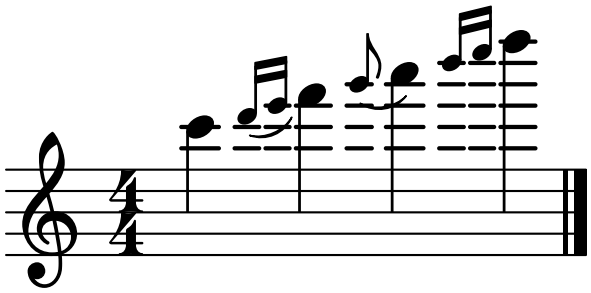


typed-scores

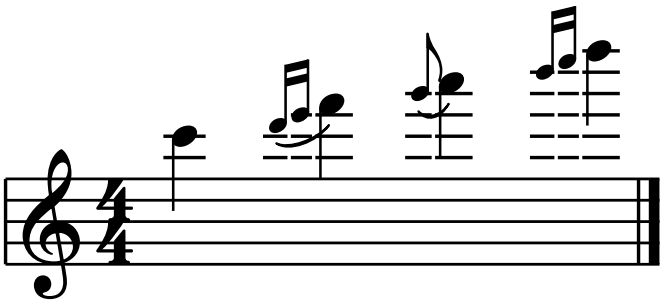


Grace-note ledger clearance

LilyPond 2.26.0



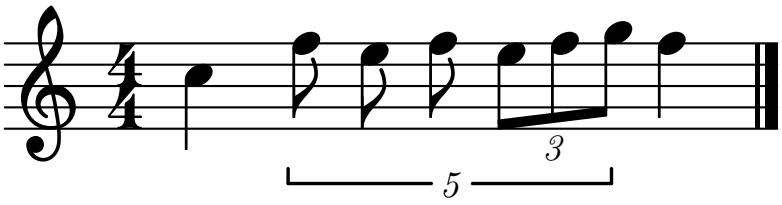
typed-scores



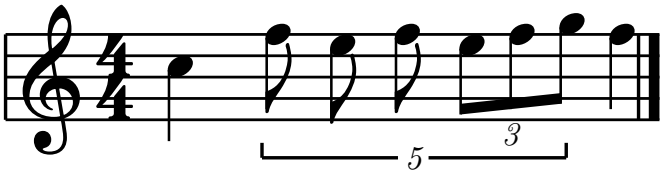
LilyPond A/B: nested tuplets

This is LilyPond's nested-tuplet example: an unbeamed outer 5:4 group with an explicitly beamed inner 3:2 group. It compares numeral size and optical centering on the note-column origins as well as bracket thickness, hook height, and beam proximity.

LilyPond 2.26.0



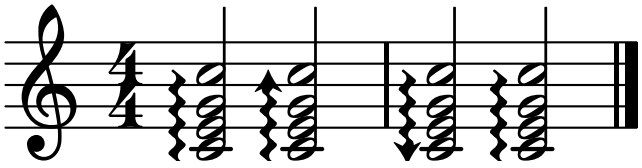
typed-scores



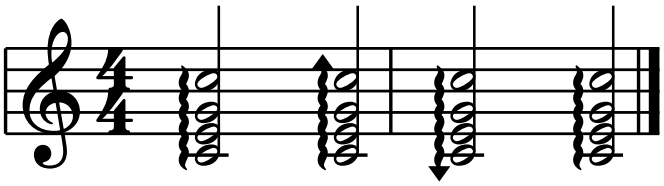
LilyPond A/B: arpeggios and tremolos

Directional arpeggios

LilyPond 2.26.0
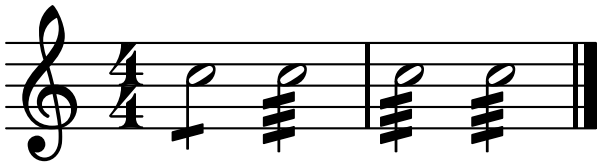


typed-scores



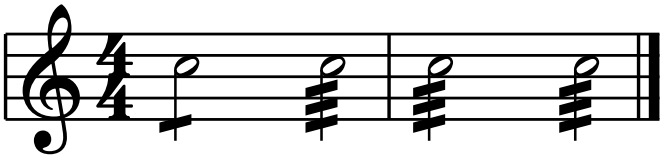
Single-note tremolo values

The comparison includes the vertical end edges and sloped faces of each StemTremolo strip, not only its overall thickness.

LilyPond 2.26.0
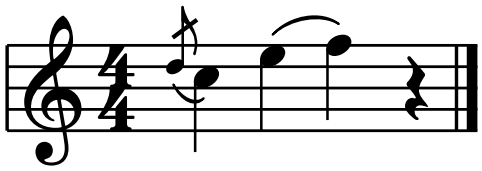


typed-scores
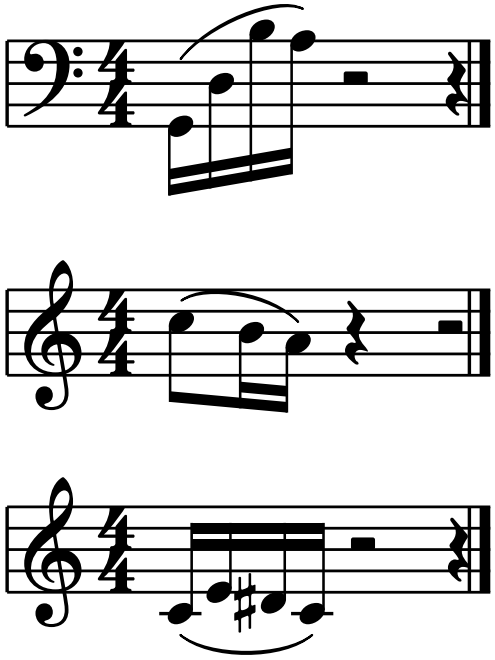


Grace slur weight

Grace-note glyphs shrink, but the independent resolving Slur grob retains the same midpoint and tapered-tip weights as an ordinary slur. Both bows below use the renderer's normal 0.22-space midpoint and 0.10-space endpoint thicknesses.



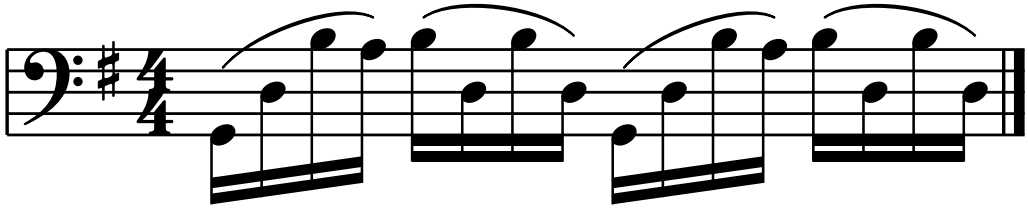
Beam-aware slur endpoints

Slur anchors and collision obstacles use the beam group's rendered stem direction. The bass phrase begins beside its low first head instead of an imaginary upward stem tip; the first treble phrase descends to its final sixteenth without lifting the right endpoint away from the note. The final treble group has upward stems throughout, so LilyPond's neutral-direction rule puts its slur below the notes.
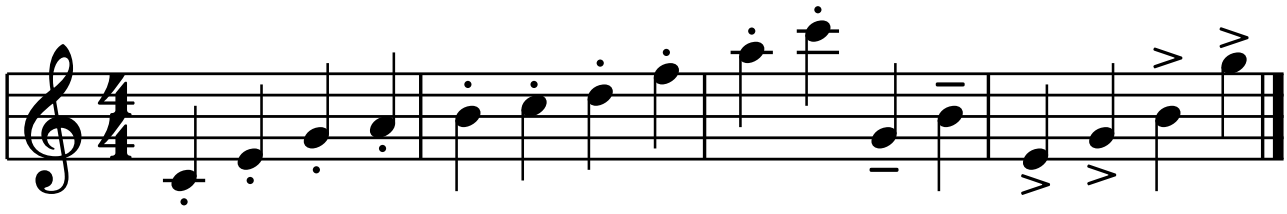


Slur clearance over beamed sixteenth runs

A slur must keep free air over every notehead it spans, measured across the head's full width rather than only above its center, and beamed boundary notes let the bow deviate an extra space from the melodic interval. In the Bach figure the second and fourth phrases arch over the returning high note and float their right tip above the final low sixteenth instead of diving onto it.



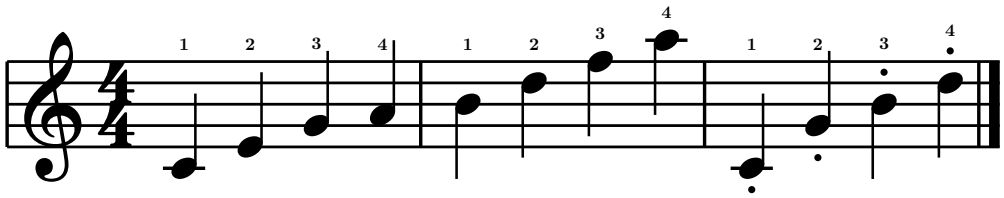
Staccato and script quantization against staff lines

Following LilyPond's script placement, a staccato dot or tenuto bar leaves its notehead and then quantizes to the half-space grid in its direction, never resting on a staff line: the dot under the G on the second line hops below the staff, the A dot centers in the first space, and dots above the middle-line B and the C skip the fourth line into the space above. Ledger notes take their dot a whole space from the pitch with no extra shift. The taller accent family always clears the staff outline by a quarter space.

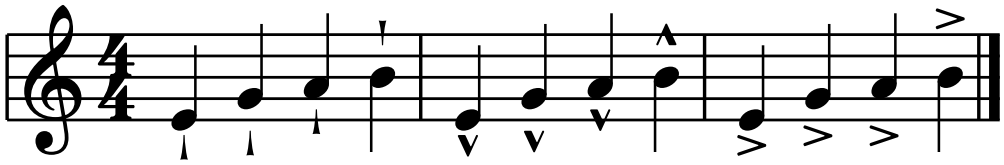


Fingering band and script family placement

Fingerings follow LilyPond's defaults: digits sit in a common band just above the staff — half a space of staff padding — and rise only when the notehead column or an articulation stacked above reaches higher; stems never push a digit up. The staccato dot below the C shares its event with a digit above.

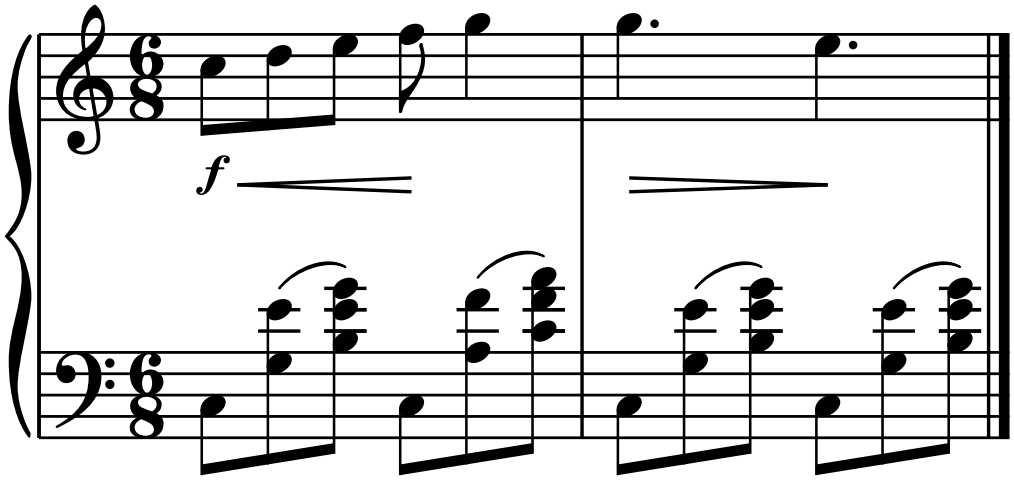


The wedge and marcato quantize their near edge to the half-space grid like the dot quantizes its center, and the accent — LilyPond's one unquantized script — clears the staff outline by a quarter space:



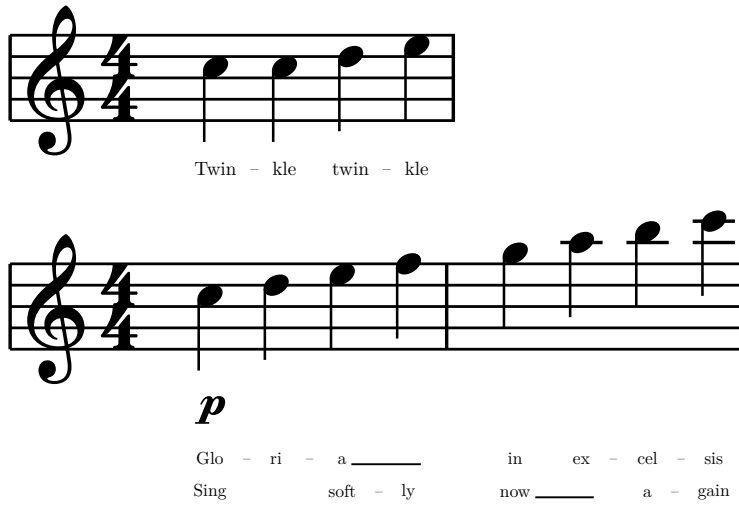
Cross-staff clearance of hairpins over slurred chords

Computed staff gaps reserve room for slur arches, so a treble-voice hairpin riding the dynamics baseline keeps free air above the bass staff's slurred chords instead of cutting through their bows.

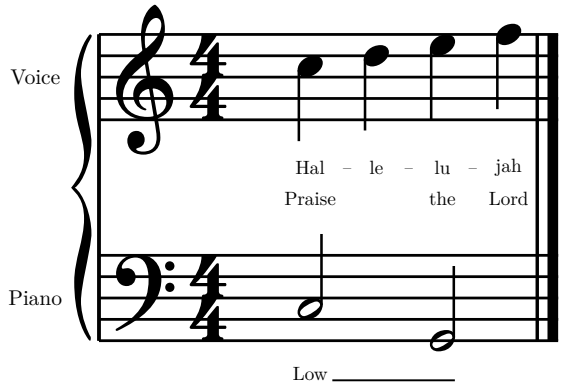


First-class lyrics

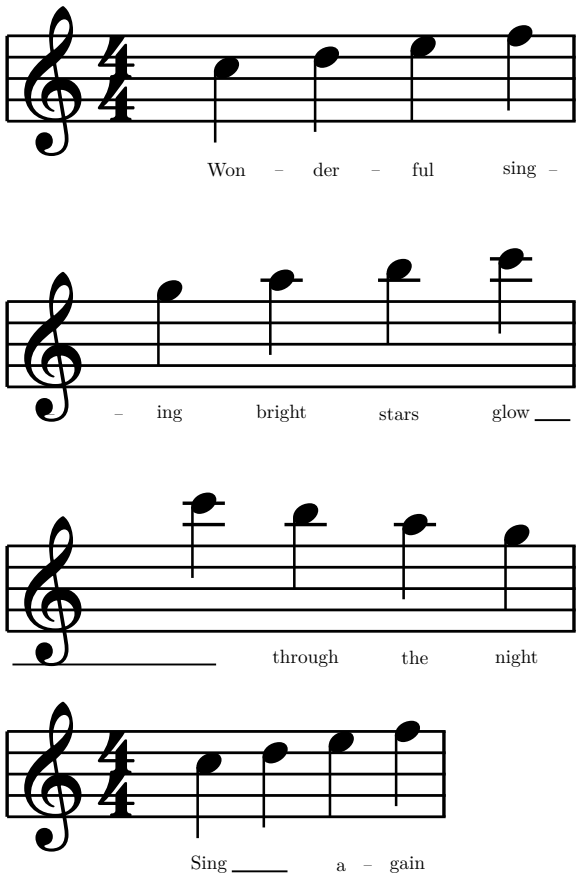
Syllables share the musical onset grid, reserve horizontal room, and use hyphens without consuming notes. Multiple verses occupy independent lanes.



Lyrics target declared staves independently and enlarge the automatic gap between the vocal and accompaniment staves.

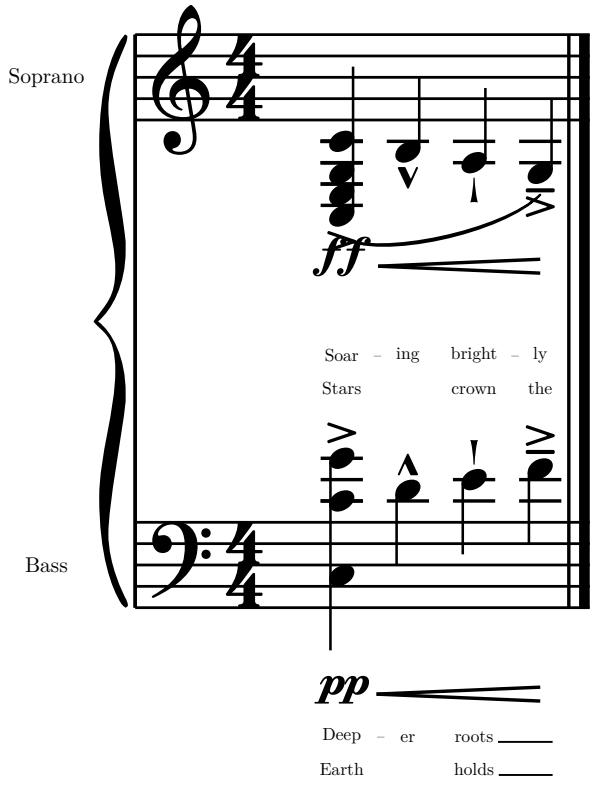


Hyphens and melisma extenders continue across barlines and wrapped systems.


Won – der – ful sing –
– ing bright stars glow
through the night
Sing a – gain

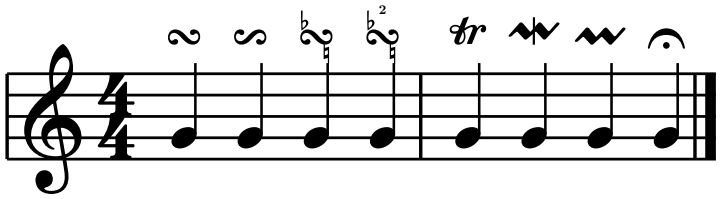
Multi-staff lyric and articulation stress test

Extreme registers, outward articulations, slurs, dynamics, direction spans, and two verses on both staves exercise the complete horizontal and vertical collision model. The automatic staff gap must clear the upper staff's lyrics from the lower staff's high ink, while the lower lyrics sit below its dynamics, hairpin, articulations, and low ledger lines.



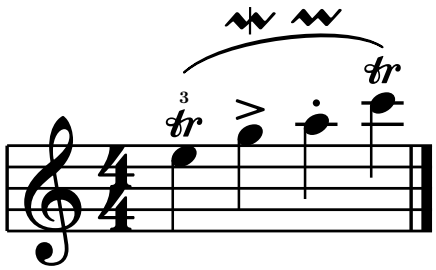
Ornament comparison

Every implemented ornament shares one script slot above the note and follows LilyPond's outside-staff stacking: the glyph clears the note ink by a fixed gap, and any fingering rides on top of it. The turn family (turn, inverted turn, and the chromatic turn with its flat and natural) uses the short glyph; the trill and the two mordents are taller and keep the same lower edge over the note.



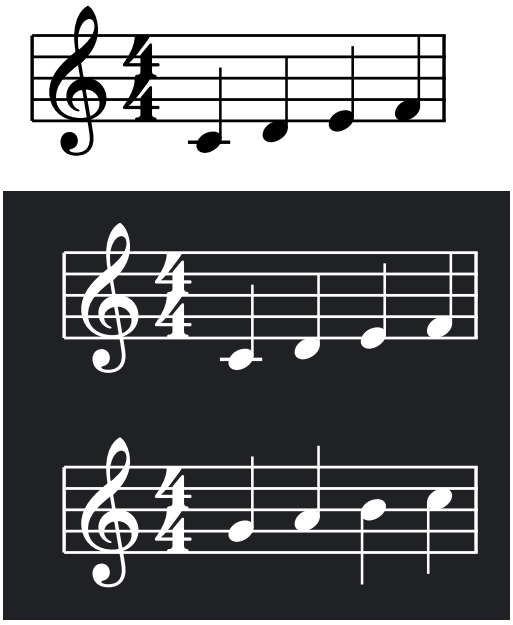
1	2	3	4	5	6	7	8
`turn`	`inverted-turn`	`chromatic-turn`	`chromatic-turn` + `turn-f=2`	`trill`	`mordent`	`inverted-mordent`	`fermata`

Ornaments stack over articulations and under fingerings, and slur bows arch clear above the whole stack just as they do over a turn:



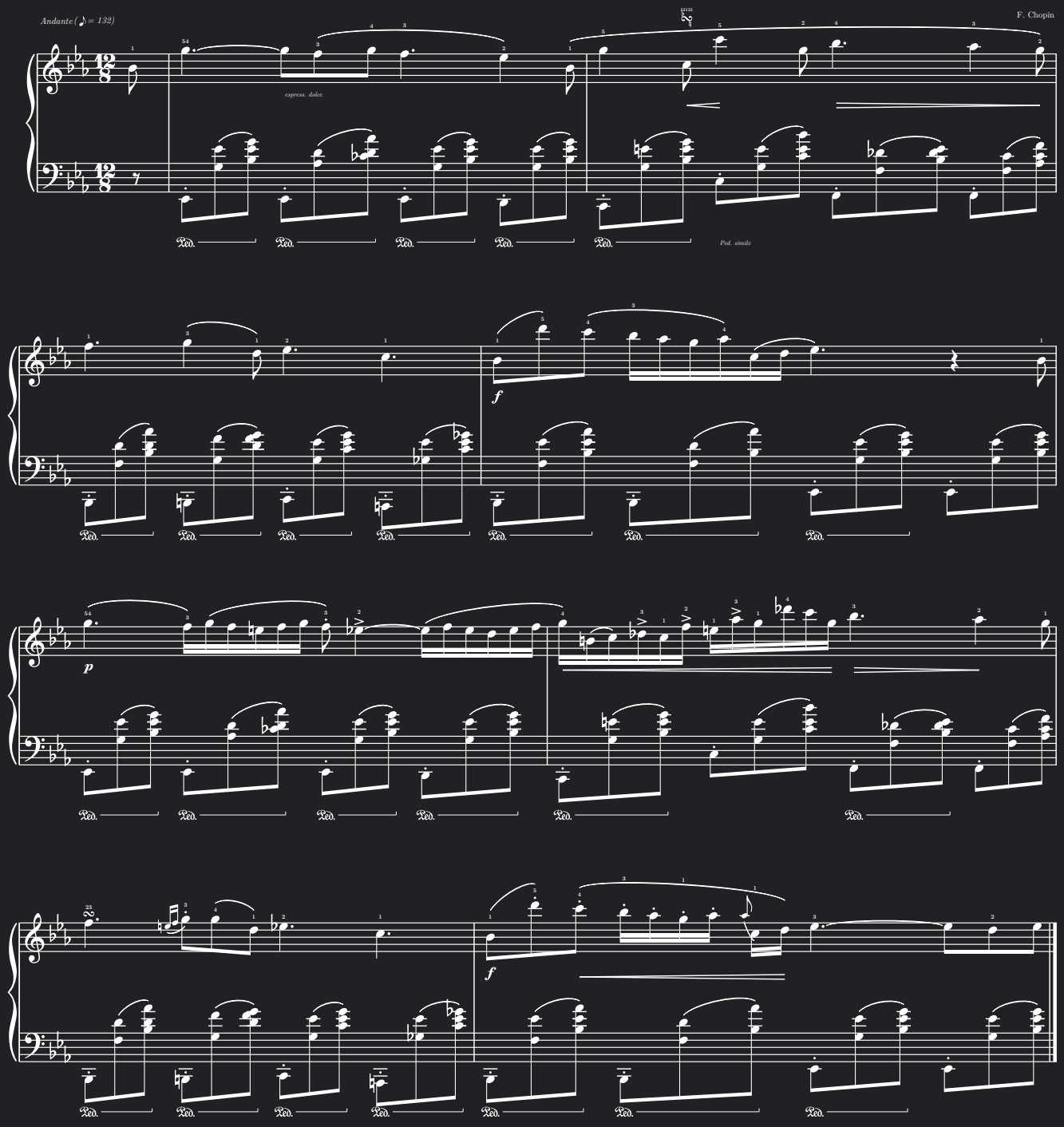
Light and dark notation themes

The light theme uses black notation ink. The dark theme uses white ink, and the automatic theme follows the surrounding solid text color. Every case recolors both CeTZ geometry and embedded Bravura glyphs.



Dark theme: Chopin Op. 9 No. 2
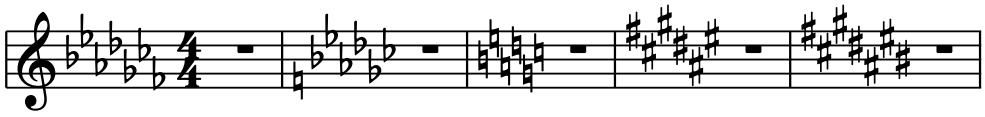
The canonical release fixture exercises a wrapped grand staff with signatures, beams, dynamics, fingerings, slurs, hairpins, pedal spans, ornaments, grace notes, articulations, and text in explicit dark mode.
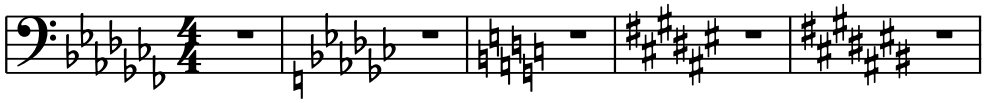


Key signature placement in every clef

Signature accidentals follow the conventional staff positions for each clef. Flats descend from the middle of the staff, and only the seventh bass flat (F flat) sits below the bottom line. Sharps stay inside the staff except for the third treble sharp (G sharp), which sits in the space above the top line. Tenor clef is the exception that cannot copy treble: its first sharp drops to the low F line so the whole signature fits within the staff.
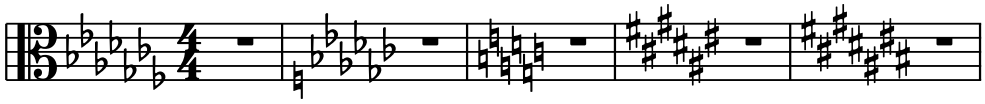
treble
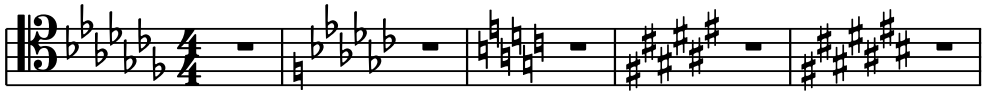


bass
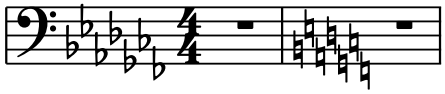


alto



tenor
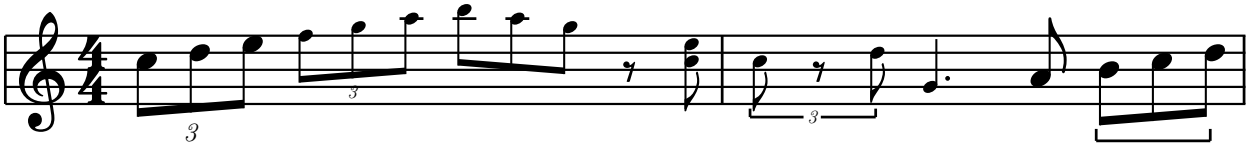


Cancellation naturals reuse the same positions as the signature they replace, so a flat key cancelled in bass clef keeps its F flat below the staff.
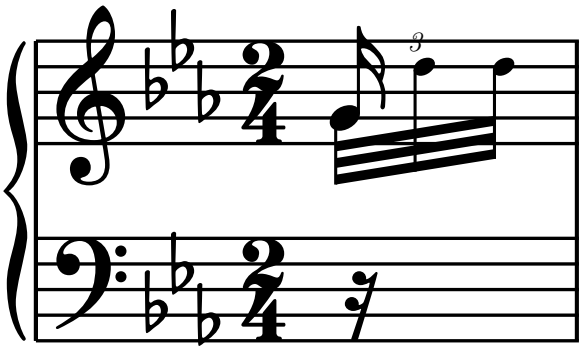


Cue-sized measured notes
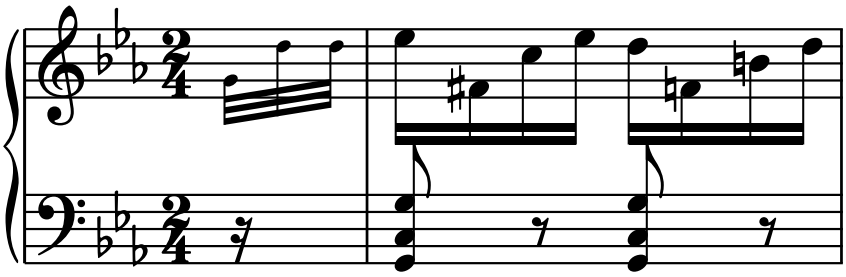
Cue groups draw notes, chords, and rests at the grace-note size while keeping their rhythmic value. The first bar compares a normal tuplet, a cue-sized tuplet with its numeral, a cue-sized tuplet with `number=never`, and a cue rest and chord. The second bar shows a bracketed cue tuplet containing a rest, a dotted cue note, and a normal-sized tuplet whose numeral is hidden while its bracket remains. Automatic beams break where cue-sized and normal-sized notes meet.



A cue group in a second voice shares a sixteenth upbeat with a normal-sized note: the normal G in voice 1 stems upward with its flag, while the cue-sized thirty-second triplet in voice 2 stems downward, merges with the coincident G notehead, and places its numeral above the noteheads without touching them.



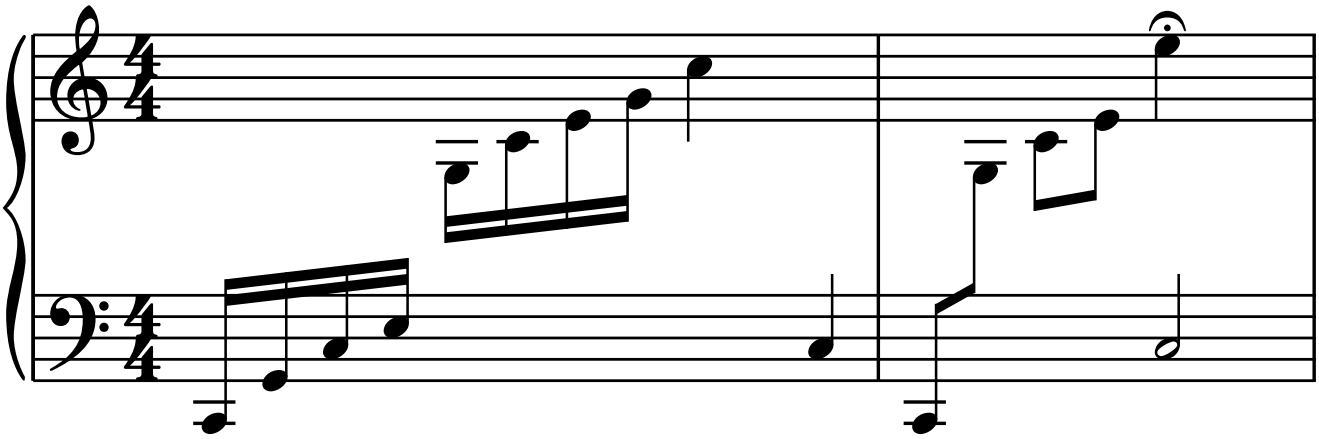
A single-voice cue-sized tuplet upbeat also fills a sixteenth pickup against a written rest in the lower staff.



Cross-staff notation

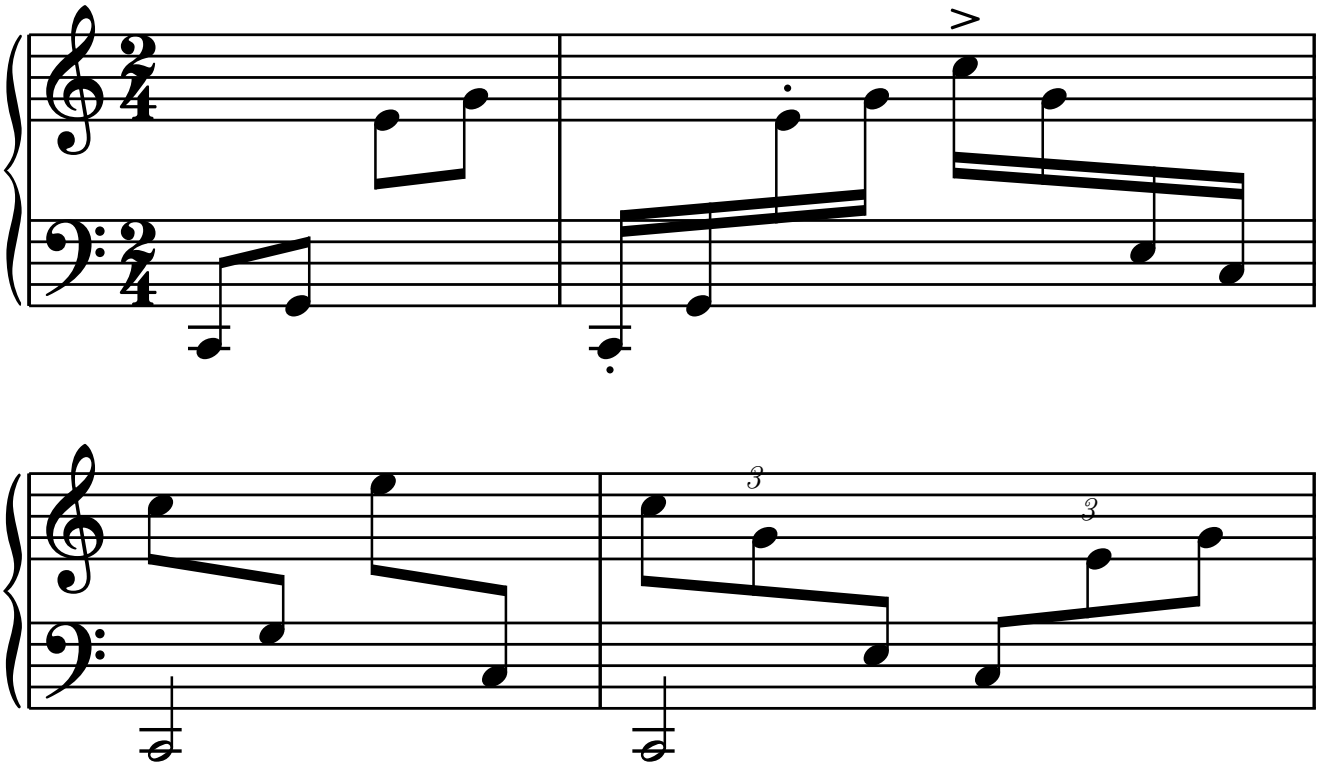
Staff switches and spacers

A lower-staff voice climbs onto the treble staff with `@upper` and returns with `@lower`. The switched notes take the treble clef, point their stems back toward the bass staff, and keep the voice's beat grouping. Spacers (`s`) keep the upper staff silent without printing rests.
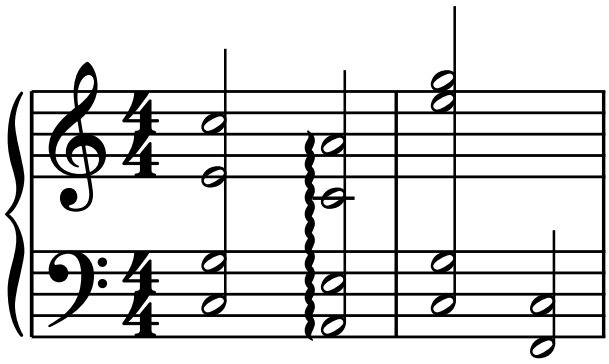


Kneed beams

Beam groups that join both staves are kneed: the beam lies between the staves, upper-staff stems point down to it, and lower-staff stems point up. Secondary beams stack toward the notes where the group begins; articulations follow the notehead side of each stem, and the staves open wide enough for the beam.



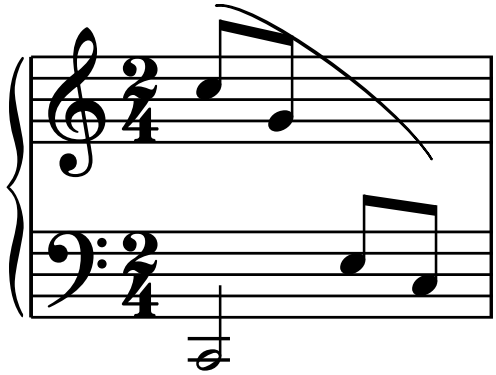
Split chords

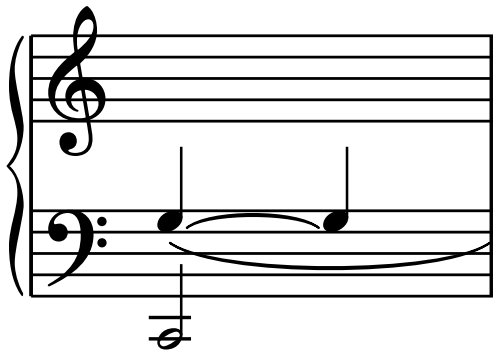
One chord spans both staves with a single stem: pitches before `@lower` or `@upper` stay on the event's staff and the rest move. An arpeggio sign follows the chord across the gap.

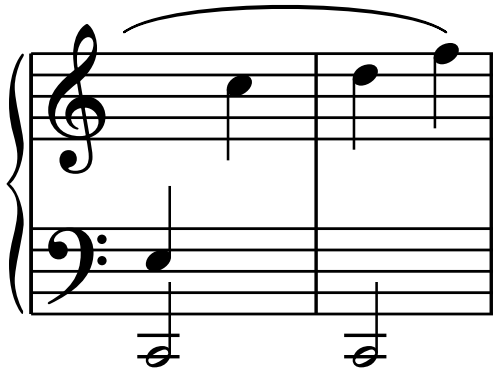


Cross-staff slurs and ties

Slurs follow a phrase from one staff to the other, including across a system break, and arch above when their notes lie on two staves. Ties are allowed while both notes are drawn on the same staff; the switch resets at every bar, so the tied target names its staff again.
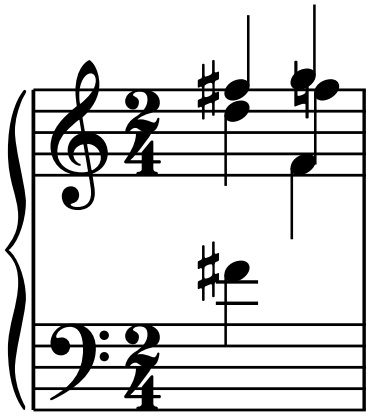






Accidentals shared by every voice on a staff
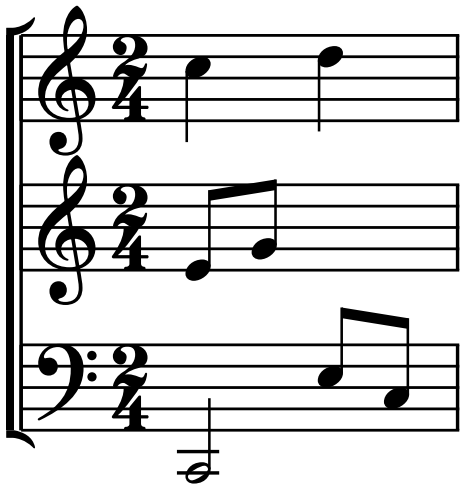
An accidental holds for the rest of the bar on the staff that draws it, across all voices. The second voice's F therefore needs a natural after the first voice's F sharp. The bass-staff voice sharpens F4 on

its own staff, so its F4 moved onto the treble staff follows the treble staff's accidentals and needs no sign.



Three staves

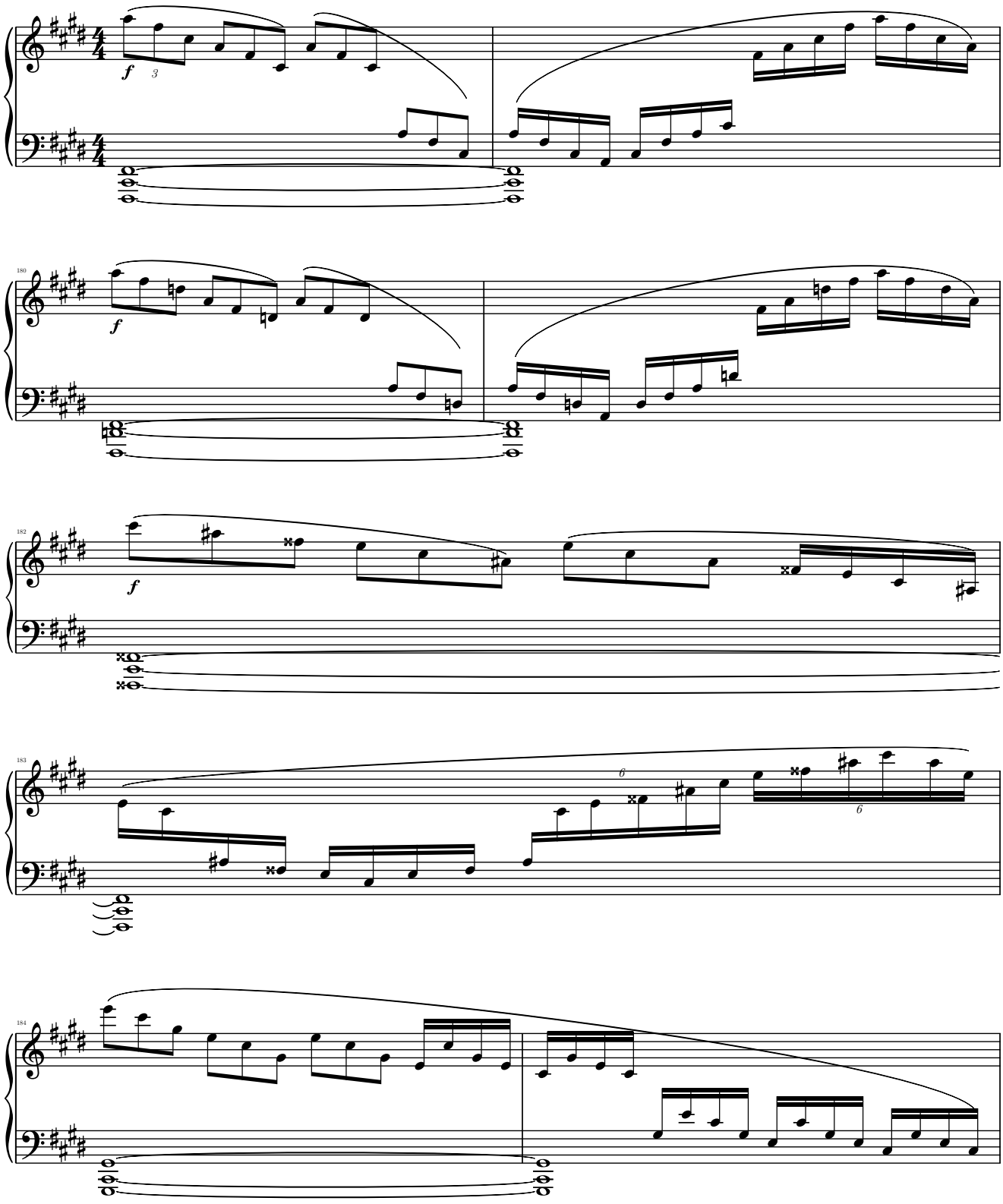
Beamed staff switches work between any neighboring staves.



Beethoven: Moonlight Sonata, measures 178–185
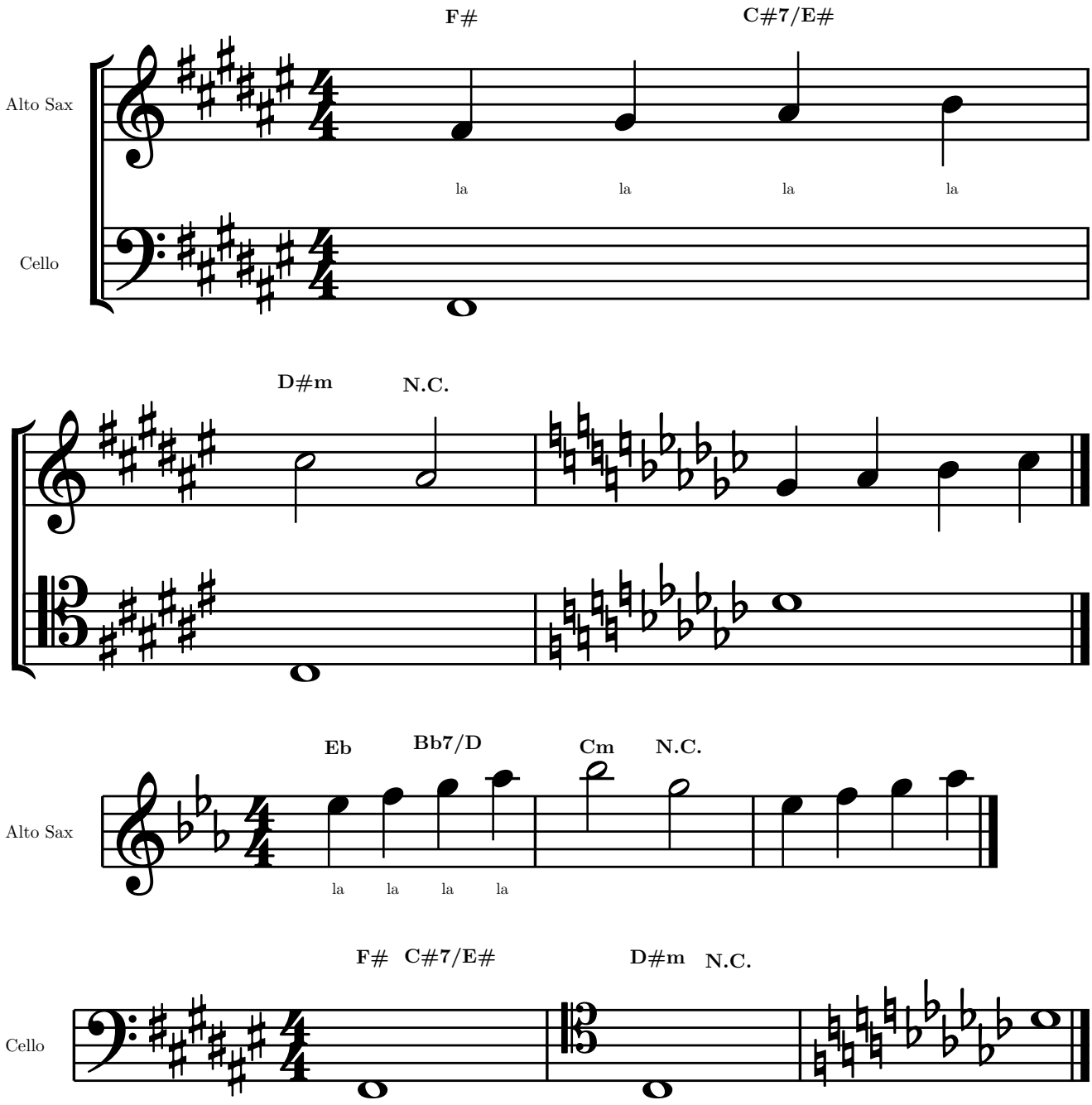
The Presto agitato coda sends the right hand's arpeggios down onto the bass staff and back up, with triplets, cross-staff slurs, and kneed sextuplet beams, all written directly in the notation language.

L. van Beethoven

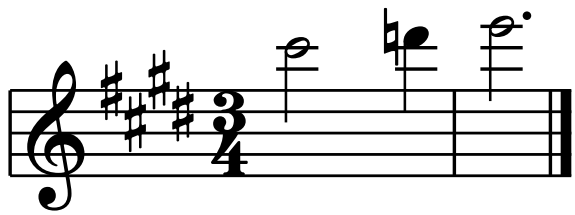


Transposition and part extraction

The duet is written at concert pitch. Its alto saxophone part moves up a major sixth: concert F-sharp major would need nine sharps, so the part is written in E-flat major and its notes and chord symbols are spelled to match. The later change to G-flat needs no new signature. The cello part keeps its clef change and lyrics-free staff, and drops the ensemble bracket and staff gap.



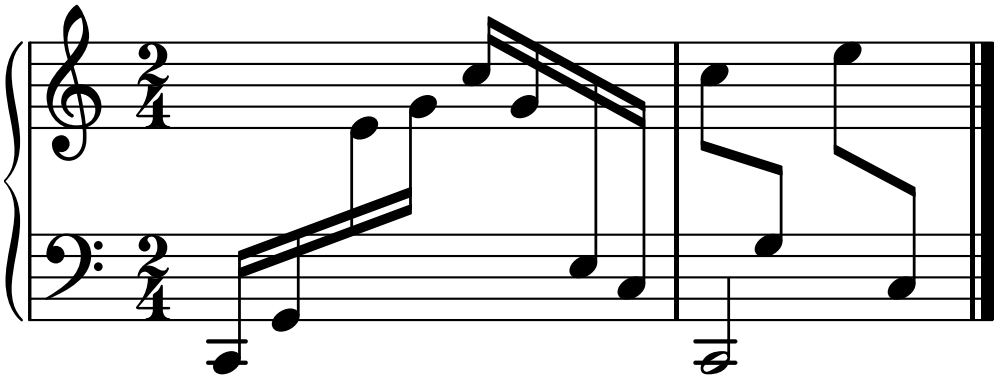
Relative octaves resolve as typed, before the interval applies: B4 to C is a step up, so a tenor saxophone part a major ninth above concert pitch still steps up from C-sharp 6 to D 6.



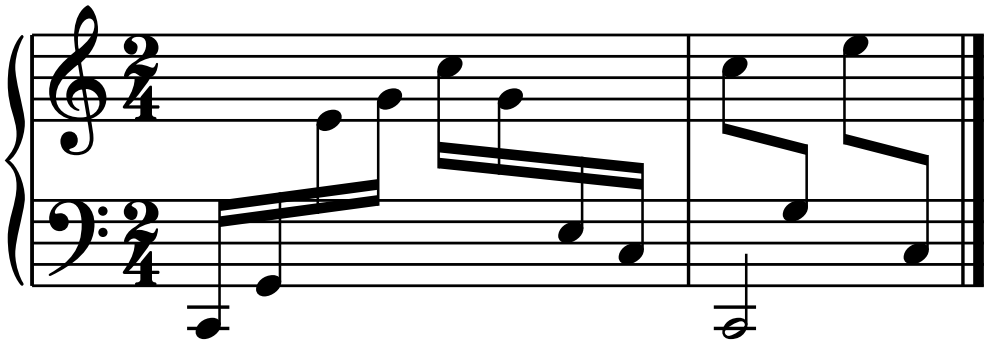
LilyPond A/B: cross-staff beams

A lower-staff voice crosses to the treble staff inside two beat groups, then an upper-staff voice alternates between the staves note by note. Both engravers knee the beam between the staves for the first group and for the alternating eighths. LilyPond keeps the second sixteenth group's stems upward under one steep beam, while `typed-scores` knees every beam group that spans two staves and limits the slant to one staff space.

LilyPond 2.26.0

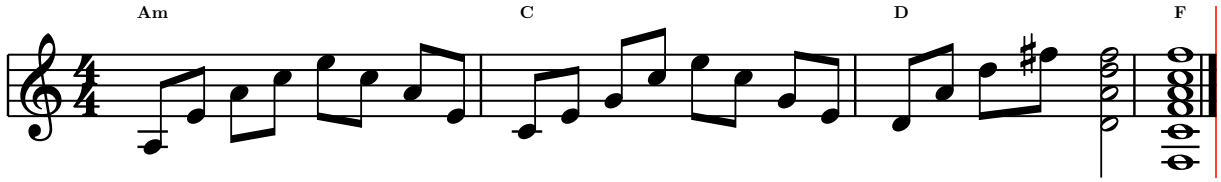


typed-scores



Compressed ragged systems stay inside their width

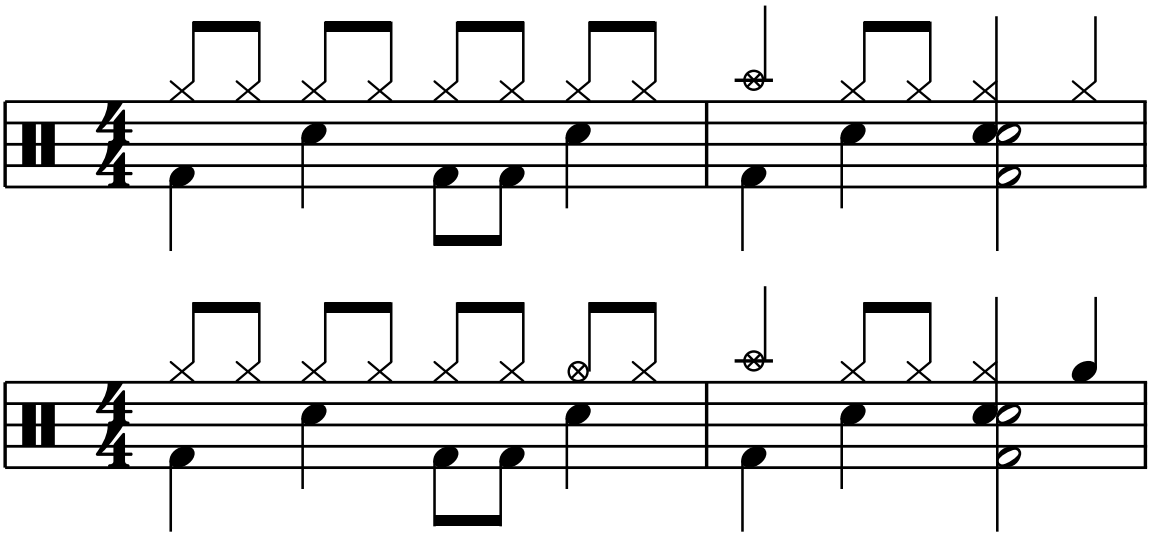
These four bars fit on one line only after the packer compresses them slightly. The single ragged system keeps that compression instead of returning to its natural width, so its final barline meets the right edge of the frame, drawn at the requested `width` of 80 staff spaces, rather than crossing it.



Percussion clef

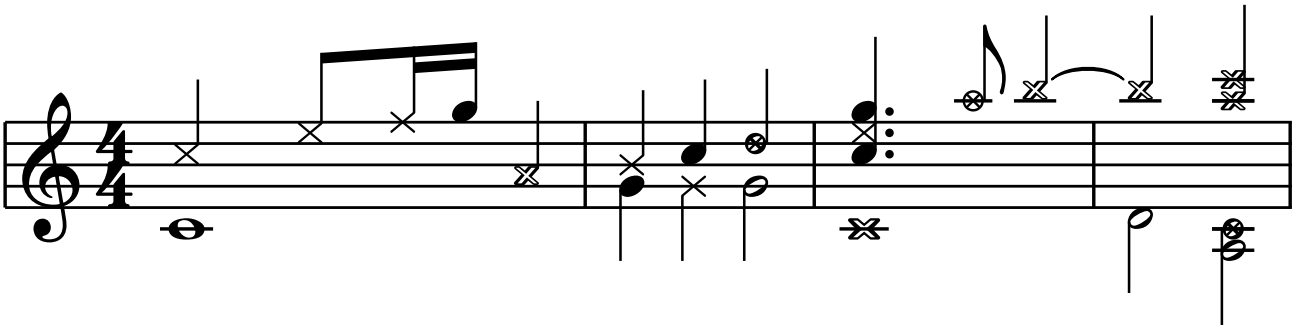
Drum-kit staff

A percussion staff reads its lines and spaces as a treble staff does. This drum kit follows the common Weinberg legend: bass drum on F4, snare on C5, hi-hat above the staff on G5, and crash cymbal on the A5 ledger line. Two voices keep the cymbals apart from the drums: the hi-hat takes X heads, the crash a circled X, and a chord marks only its hi-hat pitch. The first staff marks every head by hand; the second gives the staff a drum map and writes plain pitches, with `[circle-x]` for an open hi-hat and `[normal]` for one round head at a mapped pitch.



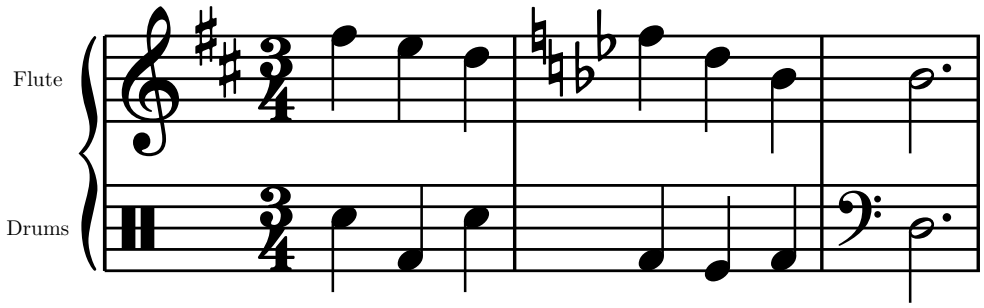
Notehead shapes

`x` after a note or chord changes its noteheads; inside a chord, `[x]` or `[circle-x]` after one pitch changes only that head. Stems attach at each glyph's SMuFL anchor while tips and beams stay where round heads put them. Half and whole notes take their own X glyphs, heads keep their ledger lines, dots, and ties, and X heads also serve spoken notes on a pitched staff. An X unison between voices does not merge with a round head.



Percussion staff in an ensemble

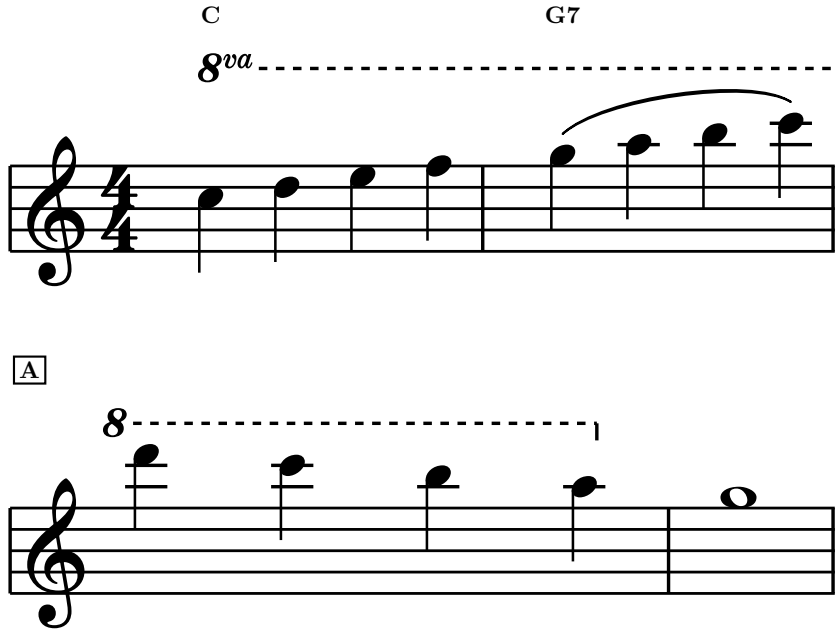
The percussion staff carries no key signature, not even at a key change, while its time signatures stay aligned with the pitched staff. Its unpitched notes never take the key's accidentals.



Ottava brackets

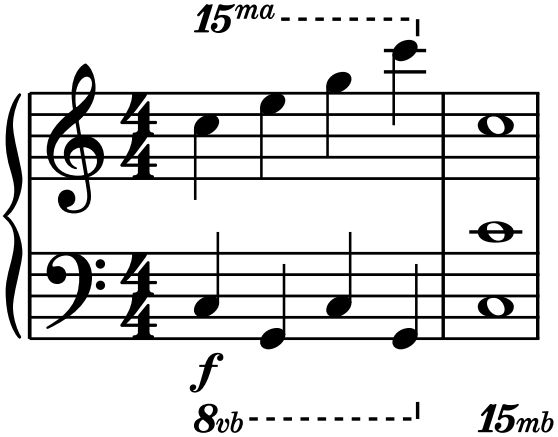
Spans across barlines and system breaks

Notes are written at sounding pitch and drawn an octave lower under `8va`. The span continues through a barline and across a system break, where the continuation shows the bare numeral; slurs and chord names stay clear.
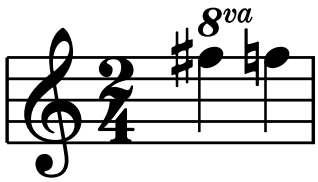


Below the staff and two octaves

`8vb` and `15mb` brackets hang below the staff under the voice's dynamics, with their hooks turned toward the notes; `15ma` shifts two octaves. A span over a single note shows the sign alone.



Accidentals under and after a bracket

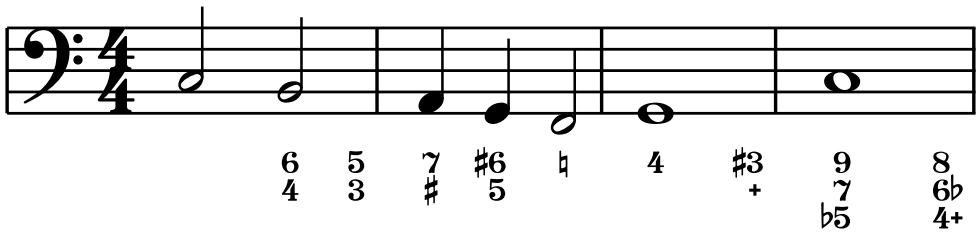
Accidentals belong to the drawn line or space, so the F natural after the span cancels the F sharp drawn on the same line inside it.



Figured bass

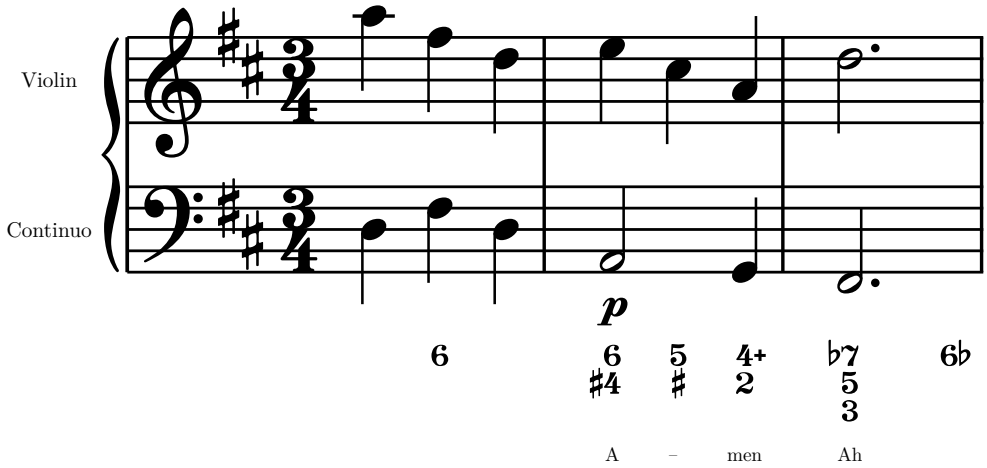
Continuo figures

Figures stack below the bass staff, centered on the note that carries them. Accidentals print before or after the numeral as written, an accidental alone alters the third, + raises a figure, and _ holds an empty row.



Figures under a continuo staff with dynamics and lyrics

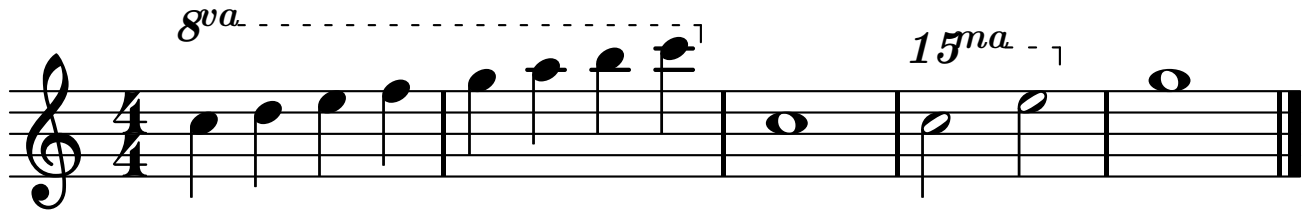
The figure band lies below the continuo staff's dynamics, and lyrics sung on that staff move below the figures. Figures that change during a held note get their own onset column.



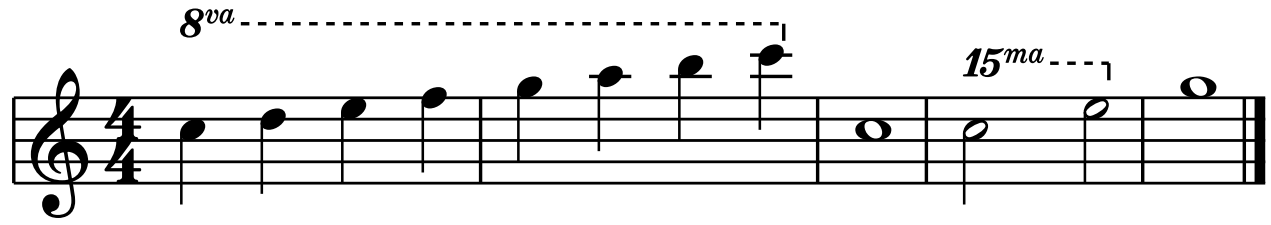
LilyPond A/B: ottava brackets

Both engravers draw the notes under `8va` an octave lower and under `15ma` two octaves lower, with dashed brackets that end in a hook at the last note. LilyPond's ordinal markups use the same text as Bravura's glyphs.

LilyPond 2.26.0



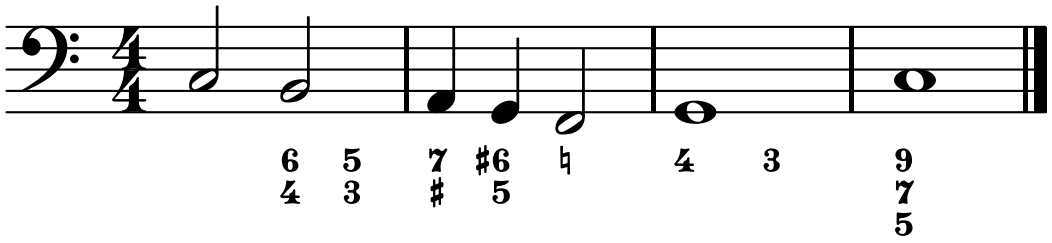
typed-scores



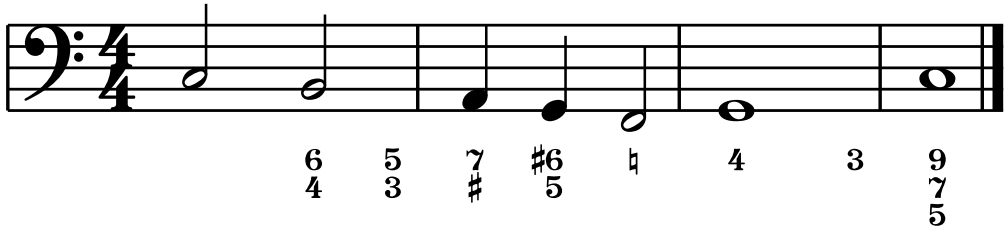
LilyPond A/B: figured bass

The same figures under the same bass line: stacks, accidentals before a numeral, an accidental alone, and a three-figure stack.

LilyPond 2.26.0



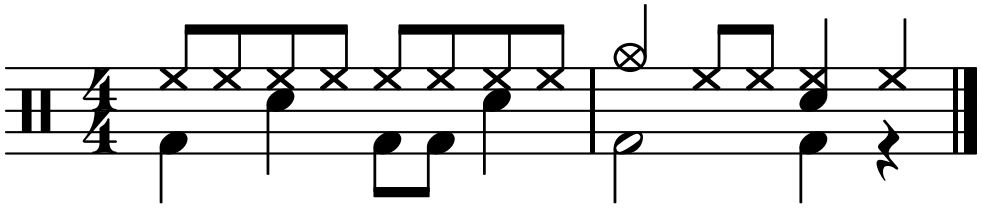
typed-scores



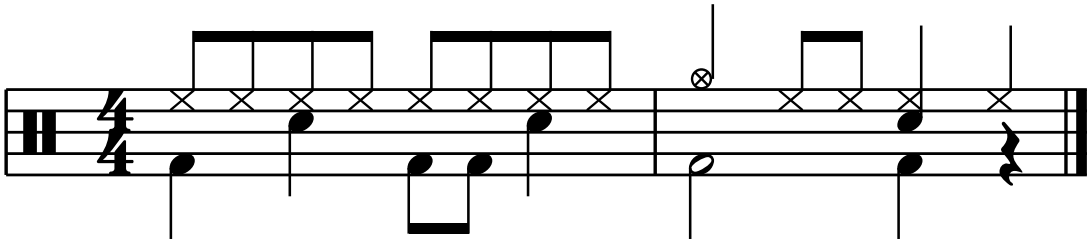
LilyPond A/B: percussion clef

LilyPond's default drum style places the bass drum on F4, the snare on C5, the hi-hat's X heads in the top space (E5), and the crash cymbal's circled X on G5 above the staff. The percussion staff reproduces that legend with the same treble positions and a drum map, `heads: (e5: "x", g5: "circle-x")`, much as LilyPond's drum style table pairs each instrument with a position and head.

LilyPond 2.26.0
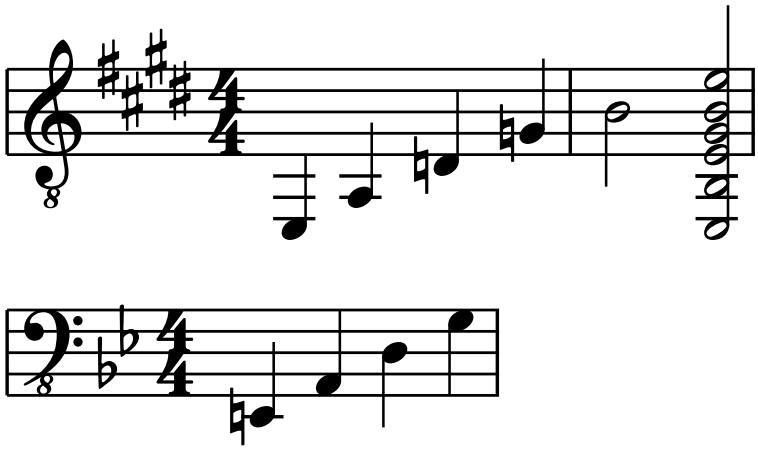


typed-scores



Guitar notation

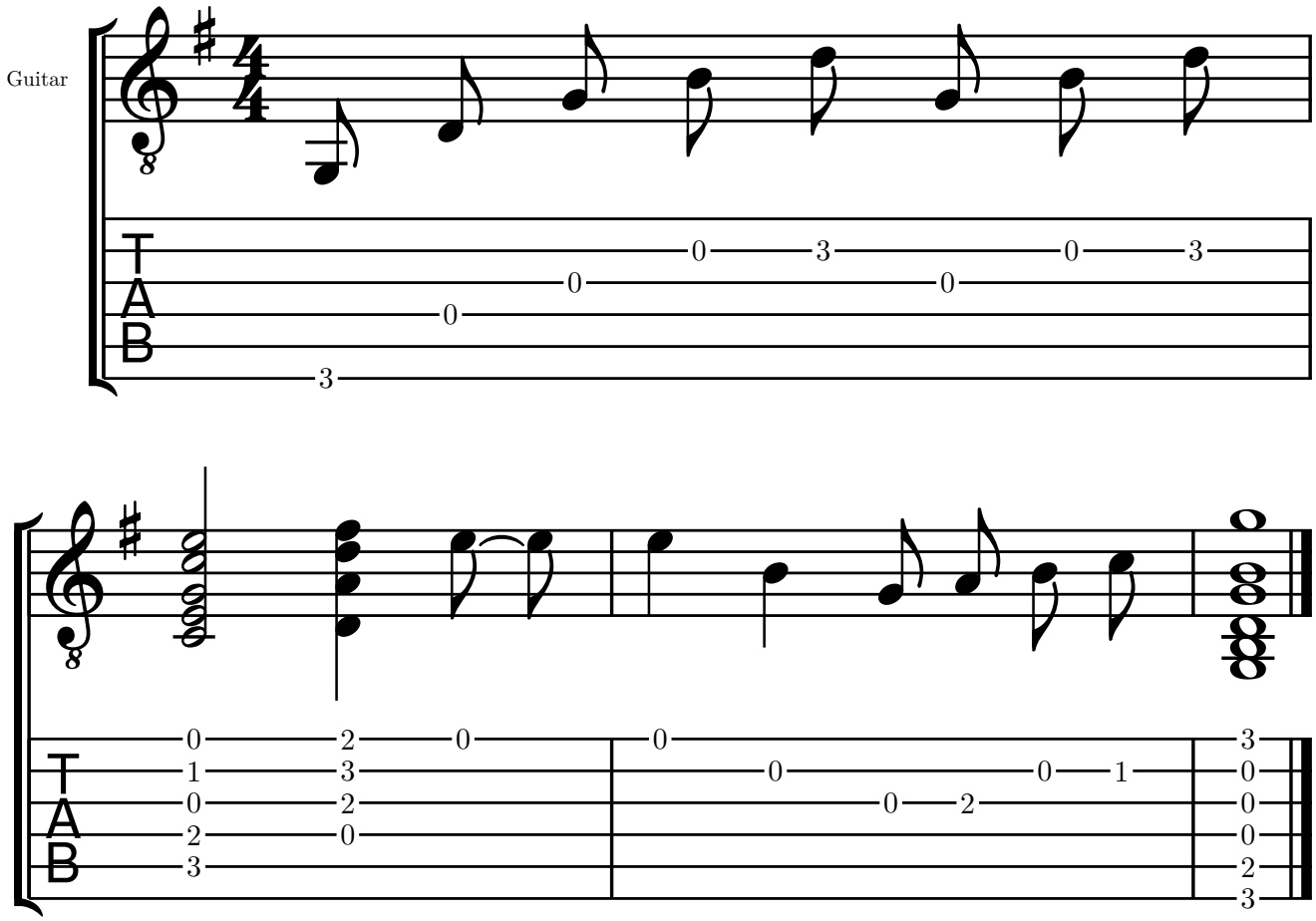
Octave clefs

Guitar and bass guitar are written an octave above their sounding pitch. `treble-8` and `bass-8` place sounding pitches on their plain clef's lines and share its key signatures, so the guitar's open strings E2–E4 sit where treble E3–E5 would.



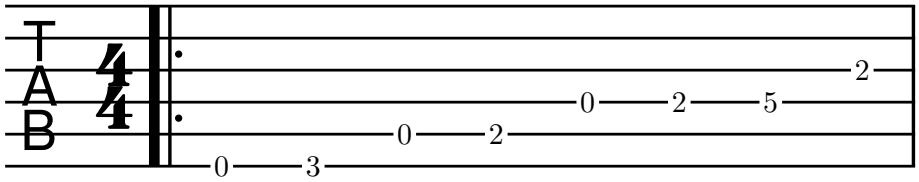
Notation with a mirrored tab staff

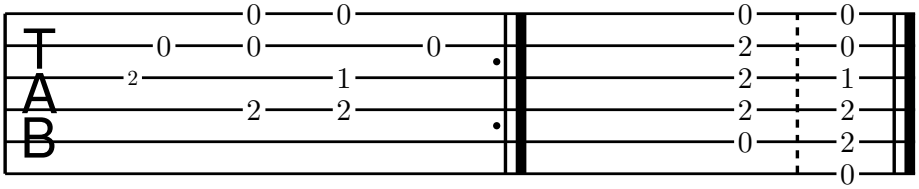
`source` repeats the guitar staff's music on the tab staff. Frets take the lowest position, chords spread across distinct strings, a tied note is not restated, and the two staves share a bracket.



Tab-only staff

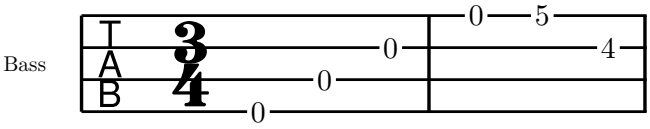
A tab staff without notation prints its time signature. `string=` chooses strings for a note or, in written order, for each pitch of a chord. Grace notes use smaller numbers, and repeat dots take the second and fourth spaces.

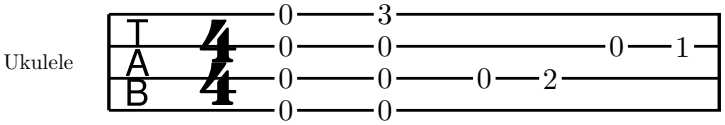


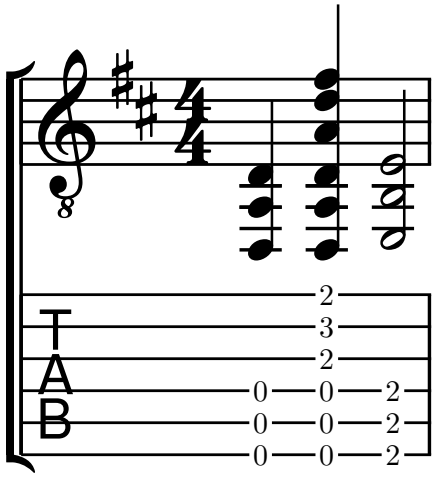


Tunings

Presets cover guitar, drop-D, DADGAD, open G, four-string bass, and re-entrant ukulele; any array of open-string pitches also works.

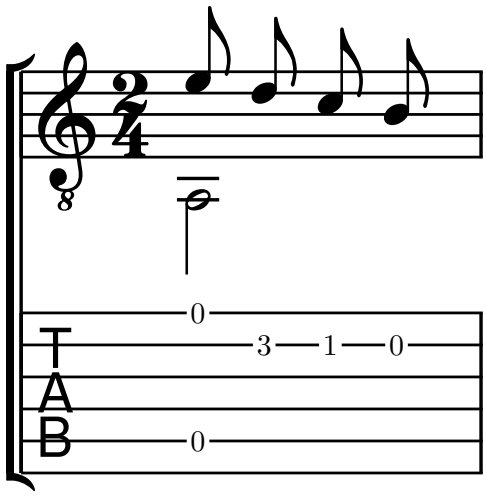


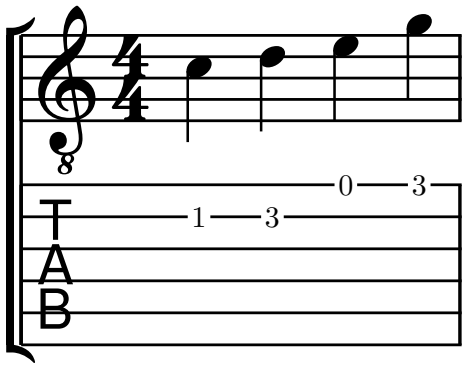


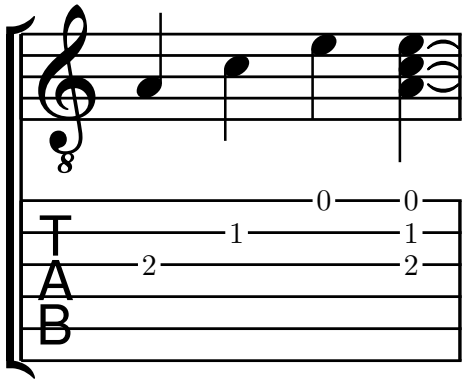


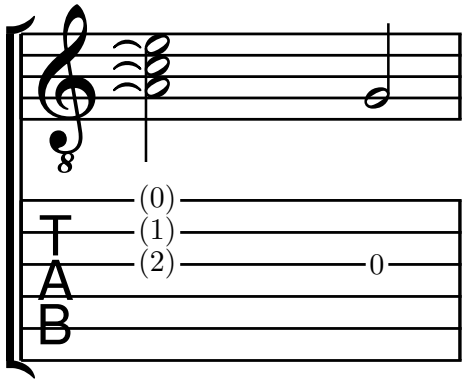
Voices share the strings

A held bass keeps its string while the melody moves above it, and a tie that crosses a line break reappears in parentheses on the new system.


0
3 1 0
0

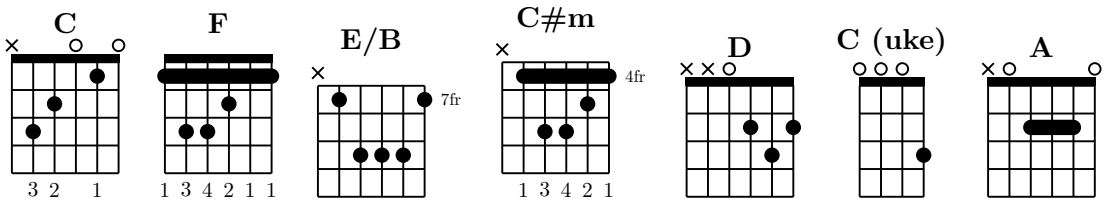

0 3
1 3


0 0
1 1
2 2


(0)
(1)
(2) 0

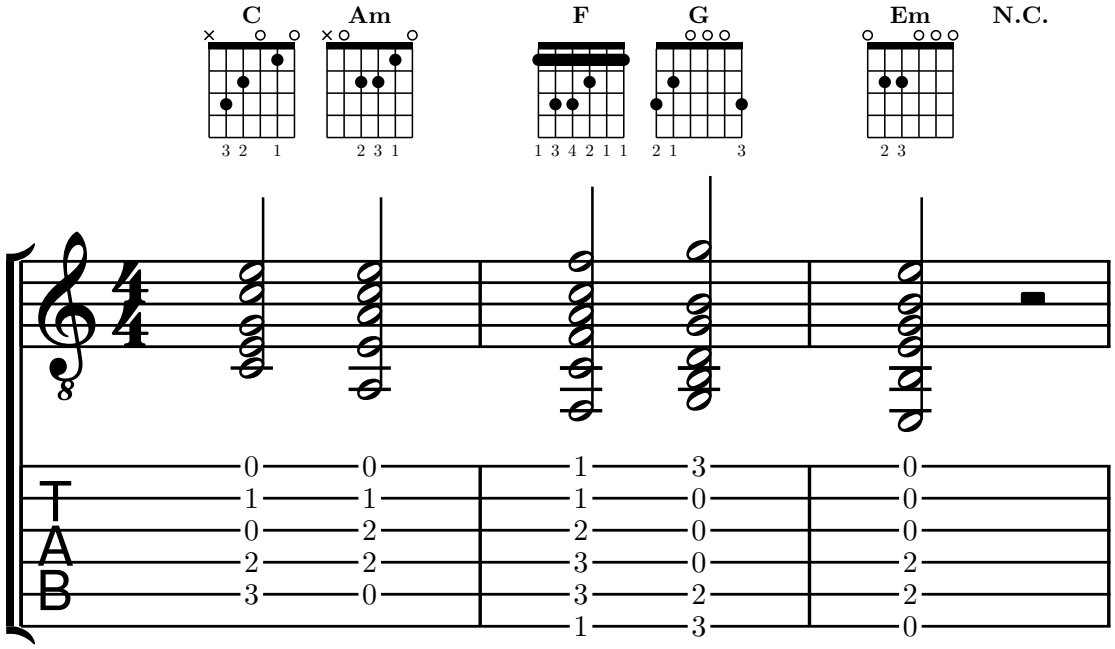
Chord diagrams

Shapes list the lowest string first. One finger across several strings at one fret draws a barre; `barre` also sets one explicitly. A shape above fret four moves the grid up with a position label.

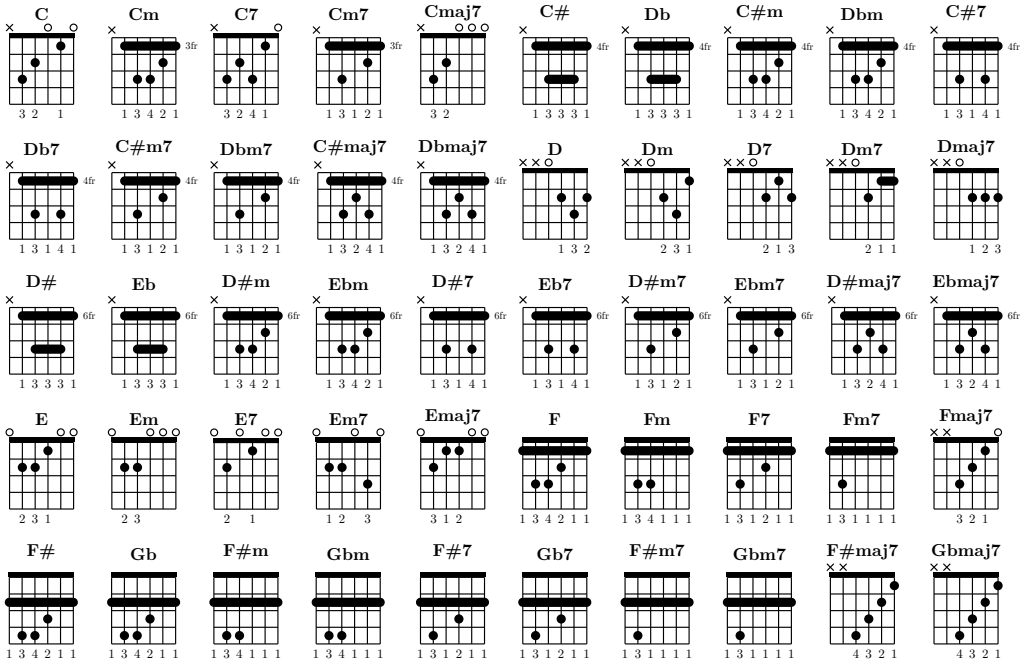


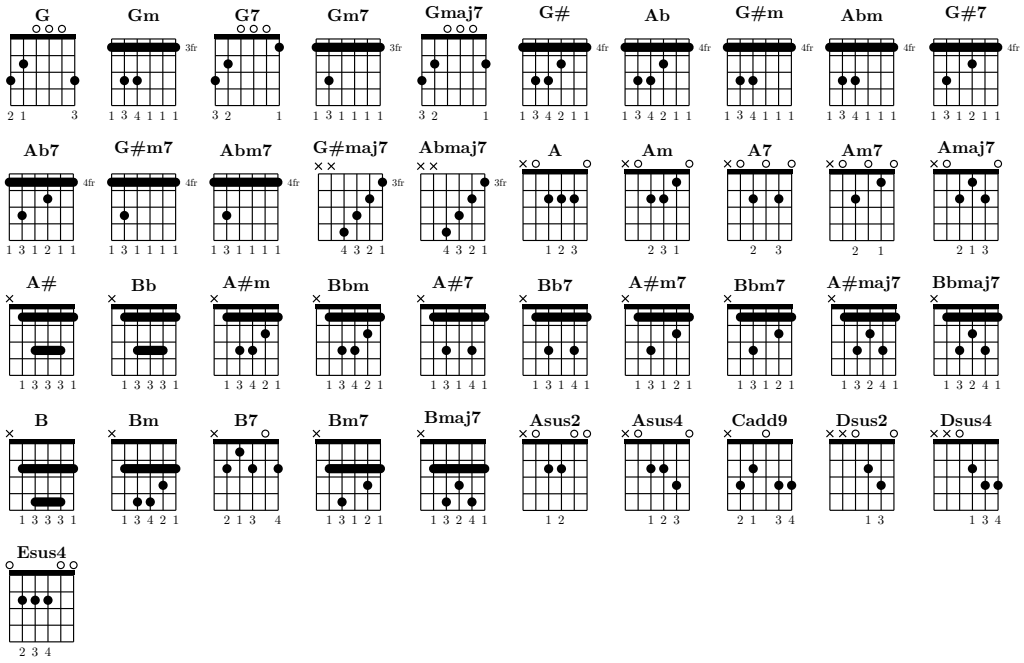
Diagrams above harmony symbols

`chord-diagrams: guitar-chords` draws the library shape for each harmony symbol above the top staff's ink, with the diagrams' tops aligned.



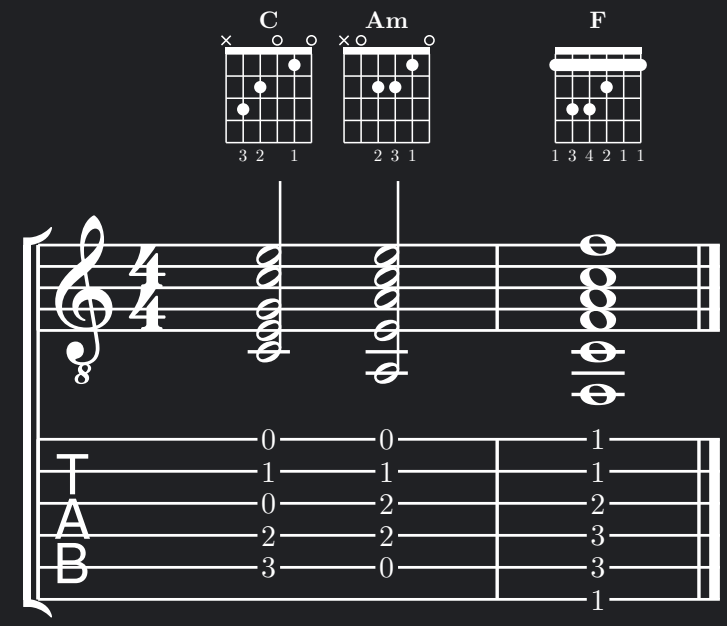
The `guitar-chords` library

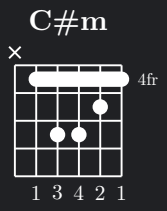



G
2 1 3
Gm
3fr
1 3 4 1 1 1
G7
3 2 1
Gm7
3fr
1 3 1 1 1 1
Gmaj7
3 2 1
G#
4fr
1 3 4 2 1 1
Ab
4fr
1 3 4 2 1 1
G#m
4fr
1 3 4 1 1 1
Abm
4fr
1 3 4 1 1 1
G#7
4fr
1 3 1 2 1 1
Ab7
4fr
1 3 1 2 1 1
G#m7
4fr
1 3 1 1 1 1
Abm7
4fr
1 3 1 1 1 1
G#maj7
3fr
4 3 2 1
Abmaj7
3fr
4 3 2 1
A
1 2 3
Am
2 3 1
A7
2 3
Am7
2 1
Amaj7
2 1 3
A#
1 3 3 3 1
Bb
1 3 3 3 1
A#m
1 3 4 2 1
Bbm
1 3 4 2 1
A#7
1 3 1 4 1
Bb7
1 3 1 4 1
A#m7
1 3 1 2 1
Bbm7
1 3 1 2 1
A#maj7
1 3 2 4 1
Bbmaj7
1 3 2 4 1
B
1 3 3 3 1
Bm
1 3 4 2 1
B7
2 1 3 4
Bm7
1 3 1 2 1
Bmaj7
1 3 2 4 1
Asus2
1 2
Asus4
1 2 3
Cadd9
2 1 3 4
Dsus2
1 3
Dsus4
1 3 4
Esus4
2 3 4

Dark theme: tab lines and diagrams

Tab lines break around fret numbers instead of masking them, so the numbers stay legible on any page color.
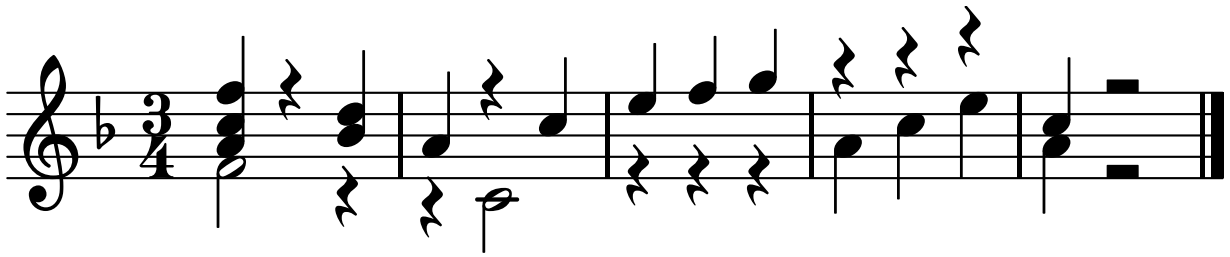




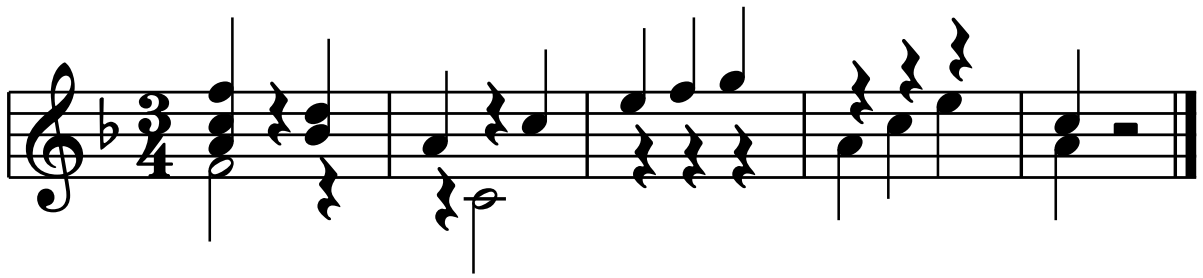
LilyPond A/B: rests between voices

A rest in one voice moves toward its own side in half staff spaces until it clears the noteheads the other voice strikes at the same onset: below the treble chords, below the A, and above the rising line. In the final bar LilyPond keeps both half rests, while `typed-scores` deliberately draws one rest for voices that rest together for the same value.

LilyPond 2.26.0
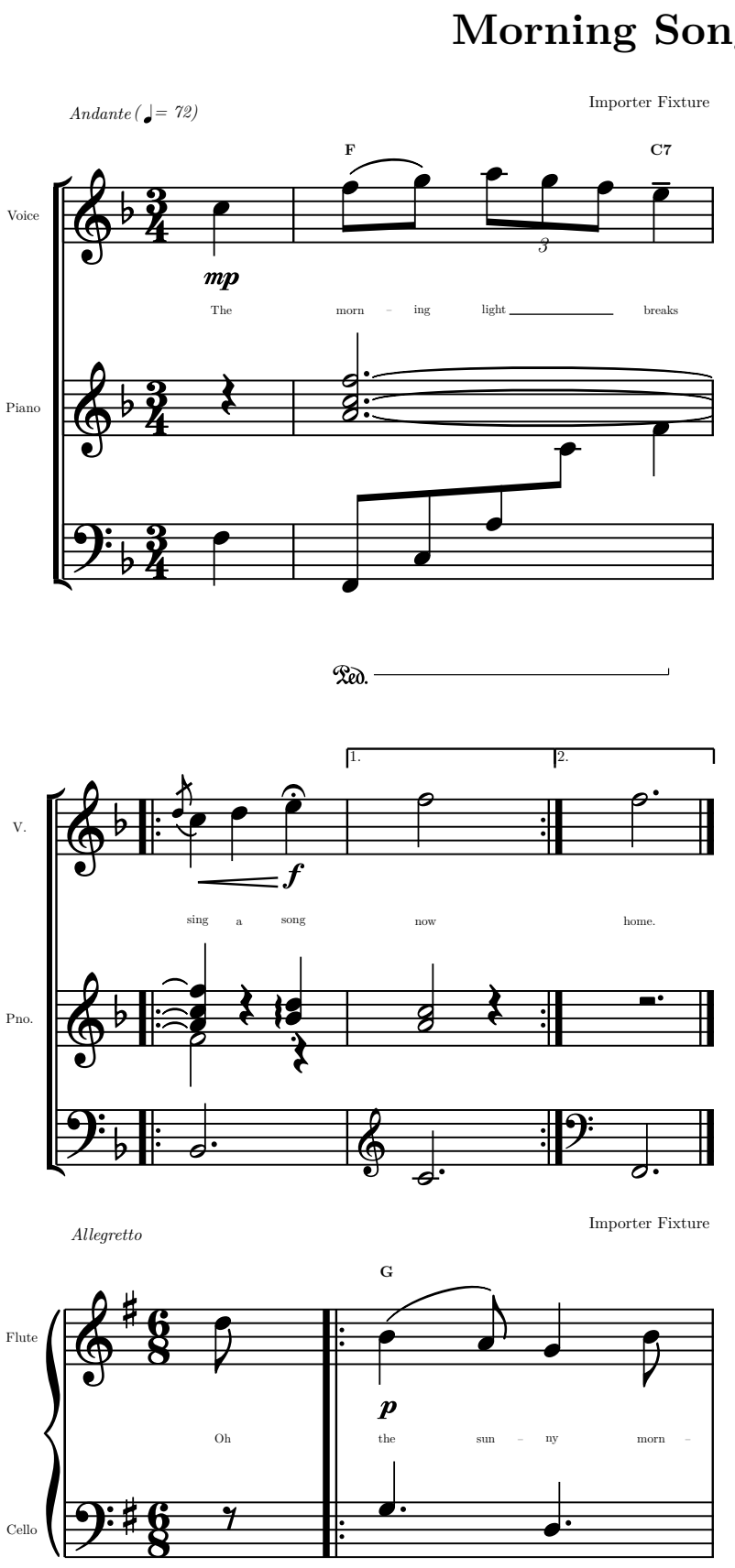


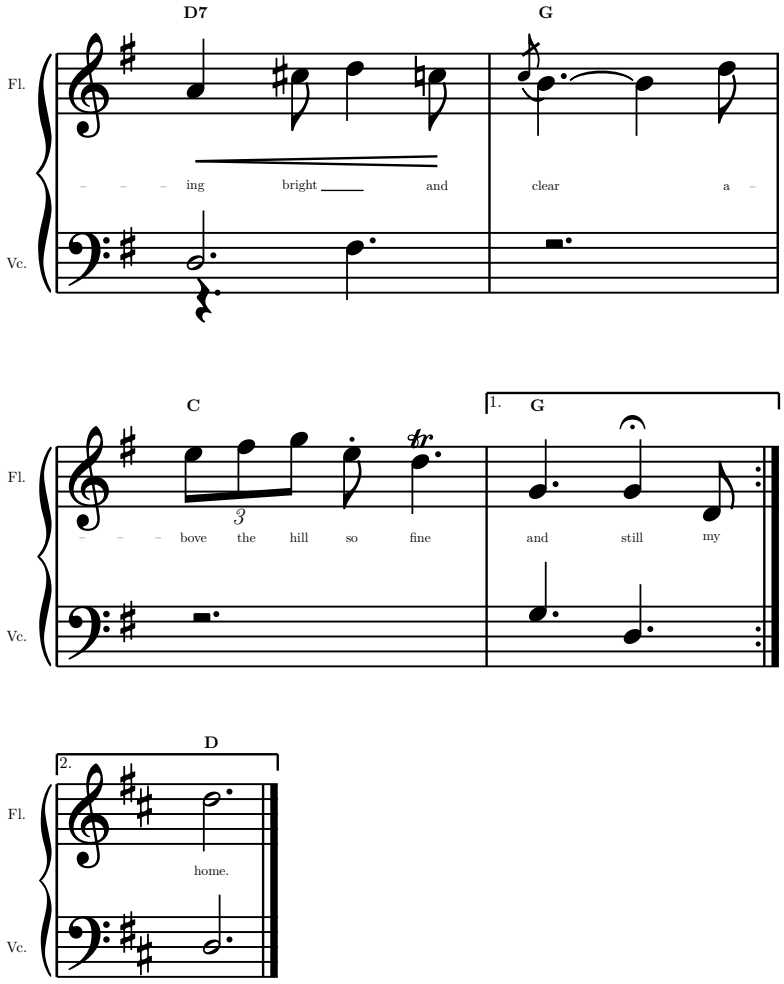
typed-scores



Importing MusicXML and ABC

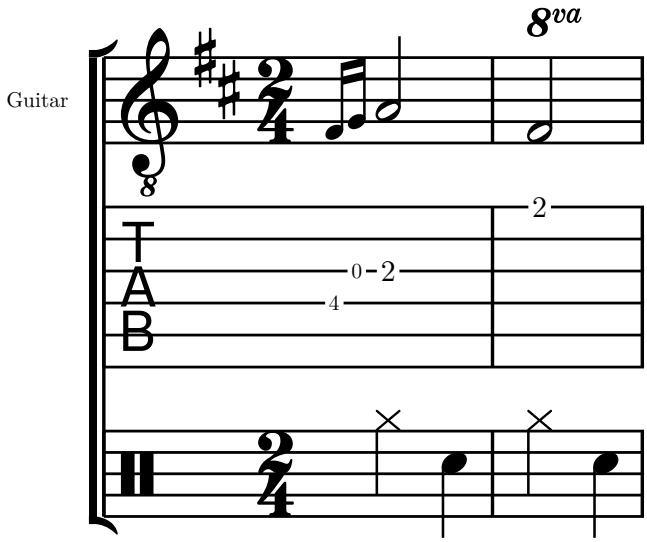
A compressed MusicXML score and an ABC tune are converted by the plugin and engraved like hand-written input: parts become labeled staves, a piano part becomes a braced pair, and repeats, voltas, lyrics, chord symbols, dynamics, pedal marks, and cross-staff beams carry over.
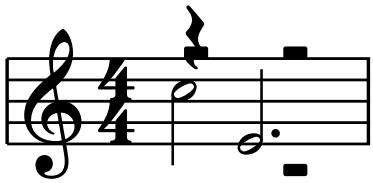



D7
G
Fl.
– – – ing bright ____ and clear a –
Vc.
C
1.
G
Fl.
– – – bove the hill so fine and still my
Vc.
D
2.
Fl.
home.
Vc.

Combined notation features

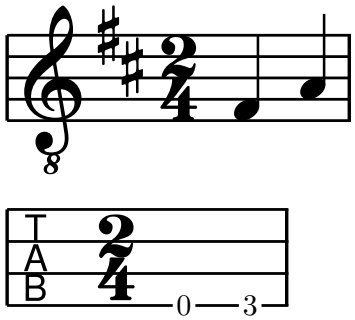
Transposition moves the pitched staff and its mirrored tab together while keeping the percussion pitches and cymbal heads in place. The ottava changes the notation's register while tab numbers retain the sounding pitches.



Rests clear notes sustained from an earlier onset. A voice's rest does not merge with another resting voice while a third voice still sustains a note.



Independent tablature can be extracted as a part, keeping its tuning and shared measure metadata. Notation parts remain usable beside mirrored tab staves in the original ensemble.



Mid-bar clef changes

Liszt's **Liebestraum No. 3** returns the left-hand staff to bass clef at each barline, then switches to treble for the two chords. The right-hand accompaniment uses beamed groups of five eighth notes. The last measure places a natural and flat before a chord after the clef change to check their spacing.

